

1 Light-sharing control for tracker-based agrivoltaics
2 under a yield floor: what learning adds and what
3 shade-response uncertainty costs

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6 **Abstract**

Single-axis trackers let agrivoltaic arrays shift the daily rotation from maximum electricity towards letting light through (light sharing). We quantify its value under a yield-retention floor of 90% of open-field yield with a differentiable simulator of the whole system. The crop response to shade is the least certain part: a hierarchical model fitted to 428 observations from 58 field studies predicts a new study of the four staple crop types only to about 0.22 in relative yield, no better than proportional loss, so we turned the learned curves and four field trials into ensembles of plausible responses by gradient descent through the simulator. On the 92 held-out crop–site pairs of the main split, light-sharing rules allowed arrays 2.0–2.9 times as dense as static arrays at equal mean retention and raised the land-equivalent ratio (LER) from 1.07 to 1.18–1.24 (1.27 for a neural controller), with a phenology rule needing no forecast the best; the floor was met in 69–76% of seasons. Pooled over three splits (76 pairs), the neural controller exceeded the written rule by 0.044 LER and a rule tuned on the same seasons by 0.018 (95% interval 0.012–0.024); a CMA-ES-optimised schedule was 0.013 behind it, and training on ensembles or a farm-measurable biomass signal did not help. Design mattered more: a density chosen under the mean response left about 70% of pairs at the floor when another plausible response held (oracle 85–87%); a floor raised by 0.02 was as good as or slightly better than an ensemble-quantile rule, and no rule was safe outside the ensemble (7–33% of pairs under three fixed responses). Under illustrative costs, light sharing breaks even with the open field at 70–92 USD MWh^{−1} and static arrays at 59, and a conventional plant is more profitable above 48. For maize and soybean (53% of the upscaled 14 PWh yr^{−1}) the bias of the simulator is uncertain.

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9 1. Introduction

10 Two of the defining challenges of this century compete for the same re-
11 source: land. The energy transition requires terawatts of new solar pho-
12 tovoltaic (PV) capacity, and ground-mounted solar farms need roughly one
13 to two hectares per megawatt. Food production must rise while crops are
14 exposed to more frequent heat waves and droughts. Warming alone is pro-
15 jected to reduce global yields of maize, wheat, rice and soybean by several
16 per cent per degree (Zhao et al., 2017), and a new generation of climate and
17 crop models shows these impacts emerging earlier than previously expected
18 (Jägermeyr et al., 2021). Croplands are also where solar power potential is
19 greatest (Adeh et al., 2019). They are flat, sunny, accessible and close to
20 rural electricity demand, so converting farmland into solar farms is tempting
21 and controversial at the same time.

22 Agrivoltaic (AV) systems combine crops and PV modules on the same
23 land and aim to resolve this conflict (Dupraz et al., 2011; Dinesh and Pearce,
24 2016; Weselek et al., 2019). By sharing sunlight between modules and plants,
25 a hectare can deliver more food plus energy than the two uses side by side,
26 with land-equivalent ratios (LER) well above one in field trials and mod-
27 elling studies (Dupraz et al., 2011; Amaducci et al., 2018; Valle et al., 2017;
28 Trommsdorff et al., 2021). The partial shade also changes the crop micro-
29 climate. It lowers canopy temperature and evaporative demand, conserves
30 soil moisture and improves water-use efficiency, especially in drylands (Mar-
31 rou et al., 2013b; Hassanpour Adeh et al., 2018; Barron-Gafford et al., 2019;
32 Elamri et al., 2018; Weselek et al., 2021). Agrivoltaics is therefore a candi-
33 date adaptation measure against heat and drought stress, in addition to a
34 mitigation measure through renewable electricity. The same shade, however,
35 also reduces the light available for photosynthesis. Regulators increasingly
36 require AV systems to retain most of the reference crop yield. The German
37 pre-standard DIN SPEC 91434, for example, requires at least 66% of the
38 reference yield (DIN, 2021), and other schemes and many farmers set stricter
39 floors. How much electricity can be produced without sacrificing food secu-
40 rity depends on the crop, the local climate, the year and the growth stage.

41 Most AV designs fix the geometry of the array, i.e. the ground coverage
42 ratio (GCR, module area per ground area) and the tilt or tracking rule, from

43 generic rules of thumb. Single-axis trackers add a degree of freedom: the
 44 modules can be rotated away from the sun to let light through when the crop
 45 needs it, as demonstrated for lettuce by Valle et al. (2017), or towards the
 46 sun to maximise electricity. Choosing the rotation every day is a sequential
 47 decision problem. The value of light depends on the crop’s phenological stage,
 48 soil water and upcoming weather, and the effect of a decision appears in the
 49 harvest months later. Recent work in this journal has used evolutionary
 50 multi-objective optimisation to lay out PV arrays on farmland (Zhang et al.,
 51 2026). Machine learning for crop management has mainly used reinforcement
 52 learning with black-box simulators (Gautron et al., 2022), which is sample-
 53 inefficient and hard to scale to global, multi-decadal assessments.

54 Two methodological strands are relevant. Differentiable process models,
 55 which couple physical simulators with learning, allow parameters and poli-
 56 cies to be fitted by gradient descent through the simulator (Reichstein et al.,
 57 2019; Shen et al., 2023), and we use them for both calibration and control.
 58 Decision making under deep uncertainty asks for designs that perform accept-
 59 ably across plausible models and not only under the best estimate (Marchau
 60 et al., 2019), and scenario-based designs have finite-sample guarantees un-
 61 der convexity assumptions that we do not meet (Calafiore and Campi, 2006;
 62 Campi and Garatti, 2008); our design rules are heuristic counterparts and
 63 we compare them with a simple safety margin.

64 Two questions follow. First, how much of the electricity that an agri-
 65 voltaic system can produce under a food-security floor comes from the array
 66 density, and how much from dynamic light-sharing control of the trackers?
 67 Second, how robust is that answer to the weakest part of any such study, the
 68 crop response to shade, and does a controller learned from a simulator add
 69 anything to a well-designed hand-written rule? Answering them requires
 70 a simulator that couples the physics of light, PV, microclimate and crop
 71 growth, a protocol in which all controllers face the same design constraints,
 72 a check of the shade response against field data, and enough sites and seasons
 73 to separate real effects from noise.

74 This paper makes three contributions. The first is a quantification of
 75 light-sharing control under a yield floor. Light sharing is parameterised by
 76 one daily level between full backtracking and the rotation that minimises the
 77 shadow on the crop, and we compare static arrays, a seasonal schedule, a rule
 78 that follows crop phenology, a rule that reacts to heat and drought stress,
 79 and a neural controller (SALS) trained through the simulator. The second
 80 is a deployable evaluation protocol. The array density of every site is chosen

from its historical seasons only, with the same rule for every method, and performance is measured on held-out sites and years, over three training seeds and three spatial splits. The third is a differentiable agrivoltaic simulator that supports both, coupling hourly solar geometry, single-axis trackers with backtracking, row-to-row shading and PV electricity, the direct and diffuse light reaching the crop, canopy temperature and reference evapotranspiration under the modules, and the SIMPLE crop model (Zhao et al., 2019) extended with a flowering-stage shade penalty. All components are differentiable tensor operations (Karniadakis et al., 2021). The PV part is benchmarked against PVGIS. The crop response to shade is learned from the open data of a meta-analysis (Laub et al., 2022), checked against field trials on rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023), and propagated through the simulator as an ensemble.

We apply the protocol to 400 crop–site pairs that represent where maize, wheat, rice, soybean and potato are grown worldwide, sampled in proportion to their harvested area (Yu et al., 2020) with local crop calendars (Jägermeyr et al., 2021), and driven by NASA POWER daily weather for 2001–2019 (Sparks, 2018) under baseline and +1.5, +2 and +3 °C climates. The main findings are as follows.

1. Light sharing is the main lever. At the same mean yield retention of 90%, arrays with light-sharing control carry 2.0 to 2.9 times the module area of static arrays and produce 1.8–2.4 times as much electricity (Section 3.1).
2. The best hand-written rule follows crop phenology and needs no weather forecast. Rules that react to heat or drought stress are not better and depend on forecast quality (Section 3.3).
3. The crop response to shade can be learned from data. A hierarchical model fitted to 428 observations from 58 field studies predicts a held-out study of the four staple crop types to about 0.22 in relative yield, no better than proportional loss (it is better only for crop types the simulator does not represent), so between-study spread is large. The simulator is close to these curves for cereals and potato but milder than the meta-analysis for maize and soybean, whereas one maize trial is milder still; the gains for those two crops are therefore uncertain and possibly overestimated (Section 3.7).
4. A learned controller adds a small gain over the best rule: 0.044 LER over the written phenology rule and 0.018 over a rule tuned on the

118 same training seasons. Training it on ensembles of shade responses, or
 119 giving it a farm-measurable biomass signal, does not help, and with
 120 the wider ensembles it did not retain its margin over the written rule
 121 in our training budget (Section 3.8).

- 122 5. The bottleneck is the uncertainty of the crop response that enters the
 123 design. A density chosen under the mean response leaves about 70% of
 124 crop–site pairs at the floor when another plausible response holds (ora-
 125 cle 85–87%). A rule built on the quantiles of an ensemble of responses
 126 raises this by 9 points for 0.04 LER, but a floor raised by 0.02 under
 127 the mean response is as good as or slightly better at equal compliance,
 128 and no rule is safe outside the ensemble, so the ensemble is useful to
 129 size a margin and not to replace it (Section 3.8).
- 130 6. Under illustrative costs, light sharing breaks even with the open field at
 131 73 to 92 USD MWh⁻¹, learning is worth about 2 kUSD ha⁻¹ yr⁻¹ over a
 132 tuned rule at 90 USD MWh⁻¹, and a conventional plant replacing the
 133 crop is more profitable at every price (Section 3.9).
- 134 7. At the yield floor, agrivoltaics on 5% of the world’s staple cropland
 135 could supply about 14 PWh yr⁻¹, but only with about 10 TWp of mod-
 136 ules, several times the installed PV capacity of the world (Section 3.11).

137 2. Materials and methods

138 2.1. Global cropland sites, weather and crop calendars

139 We studied five staple crops that together cover about 670 Mha of har-
 140 vested area and 580 Mha of physical cropland: wheat, rice, maize, soybean
 141 and potato. Harvested areas (rainfed and irrigated) were taken from the
 142 SPAM 2010 gridded production maps (Yu et al., 2020) and aggregated to 1°
 143 cells. For each crop, 80 cells with at least 5000 ha of the crop were drawn
 144 without replacement with probability proportional to harvested area. An un-
 145 weighted average over the sample therefore approximates an area-weighted
 146 average over global production. The 400 crop–site pairs lie in 381 distinct
 147 cells in 50 countries (Fig. 2). Sowing and maturity dates came from the
 148 GGCM Phase 3 crop calendar at 0.5° (Jägermeyr et al., 2021). For wheat
 149 we used the winter-wheat calendar unless the coldest-month mean temper-
 150 ature was below −8°C, in which case the spring-wheat calendar was used.
 151 Daily weather for 2001–2020 was retrieved from NASA POWER (Sparks,
 152 2018): global and diffuse shortwave irradiance, maximum, minimum and

dew-point temperature, precipitation and wind speed at 2 m. Each site contributes 19 seasons (sowing years 2001–2019), each simulated as the 365 days after sowing, which gives 7600 site-seasons.

Climate change was represented by the global warming levels +1.5, +2 and +3 °C relative to 2001–2020. Daily maximum, minimum and dew-point temperatures were shifted uniformly, which keeps relative humidity approximately constant and raises the vapour-pressure deficit. The atmospheric CO₂ concentration was set to 390, 450, 500 and 600 ppm for the baseline and the three warming levels. Precipitation and irradiance were left unchanged. This delta approach isolates the thermal signal of warming; its limitations are discussed in Section 4.5.

2.2. A differentiable agrivoltaic simulator

The simulator advances all site-seasons in parallel with a daily time step for the crop and an hourly time step for radiation and PV (Fig. 1). All equations are implemented as PyTorch tensor operations (Paszke et al., 2019), so that yields and electricity are differentiable with respect to the design and control variables.

Hourly irradiance. Daily global and diffuse irradiance were disaggregated into hourly values with the Collares-Pereira and Rabl (1979) and Liu and Jordan (1960) ratios and renormalised to the daily totals. Beam irradiance on the horizontal is the difference between the two. Solar position follows the standard declination and hour-angle equations.

Array geometry and PV electricity. We consider elevated horizontal single-axis trackers with north–south axes. The axis height is 1.25 module widths, e.g. 2.75 m for the common 2.2-m single-module-high (1P) tracker. The rotation ϕ is limited to $\pm 60^\circ$. The ground coverage ratio g is the module width divided by the row pitch and is restricted to $g \leq 0.5$. For 2.2-m modules this keeps a pitch of at least 4.4 m, and a free strip of at least 2.2 m between the rows when the modules are horizontal, which is the stow position during field operations. Let ψ be the solar zenith angle projected onto the east–west vertical plane. The energy-optimal rotation with backtracking (Lorenzo et al., 2011) is $\phi_{\text{bt}} = \psi - \text{sgn}(\psi) \arccos(\cos \psi / g)$ when $\cos \psi < g$, and $\phi_{\text{bt}} = \psi$ otherwise. The *light-sharing* rotation ϕ_{sh} is the feasible angle closest to edge-on to the sun, which minimises the shadow on the crop. Given a light-sharing level $u \in [0, 1]$ chosen once per day, the rotation in every hour is

$$\phi = (1 - u) \phi_{\text{bt}} + u \phi_{\text{sh}}. \quad (1)$$

Plane-of-array irradiance combines three terms: beam irradiance on the module, reduced by the fraction of the module shaded by the neighbouring row, $\max\{0, 1 - \cos \psi / (g |\cos(\phi - \psi)|)\}$; isotropic sky diffuse; and ground-reflected light (albedo 0.2). The cell temperature follows the NOCT model (45 °C), with an hourly air temperature interpolated sinusoidally between the daily extremes. Electricity per unit land is plane-of-array irradiance $\times \eta \times g$, with $\eta = 0.21 [1 - 0.0035(T_{\text{cell}} - 25)] (1 - 0.14)$ for module efficiency, temperature losses and system losses. The installed capacity is $2.1 g \text{ MWp ha}^{-1}$. A conventional ground-mounted PV plant ($g = 0.4$, backtracking, no crop) is the energy reference E_{PV} .

Validation of the PV model. We compared the simulated specific yield of an isolated tracker row ($g = 0.05$, negligible row-to-row shading) with PVGIS 5.2 (European Commission, Joint Research Centre, 2022), for a horizontal north-south single-axis tracker with 14% losses, at 30 randomly chosen sites (29 returned data). Across sites, the simulated and PVGIS specific yields were correlated with $r = 0.971$. The simulator was on average 4.9% lower (mean absolute deviation 5.5%). This is consistent with its $\pm 60^\circ$ rotation limit and with the different irradiance databases (NASA POWER vs. SARAH2, ERA5 and NSRDB). Electricity estimates are therefore, if anything, conservative.

Light and microclimate at the crop. Averaged over the row pitch, the fraction of ground reached by beam light is $1 - \min\{1, g |\cos(\phi - \psi)| / \cos \psi\}$. The fraction of sky diffuse reaching the ground is a two-dimensional isotropic sky-view factor. We pre-computed it for 25 values of g and 13 rotation angles by ray casting over 31 neighbouring rows, and interpolate it bilinearly (differentiably). The daily ratio s of the irradiance at the crop to that of the open field defines the crop's shading. The canopy maximum temperature is lowered by $\kappa(1 - s)$.

κ is uncertain. Field studies under PV modules report small reductions of daytime crop or canopy temperature and little change in air temperature (Marrou et al., 2013b; Barron-Gafford et al., 2019), and heat-stress thresholds calibrated on air temperature do not transfer directly to canopy temperature (Siebert et al., 2014; Webber et al., 2018). We therefore use a moderate $\kappa = 1^\circ\text{C}$ as the main case and report $\kappa = 0$ (no cooling) and $\kappa = 2^\circ\text{C}$ for all methods (Section 3.2). Nocturnal warming under the modules is not represented. The reference evapotranspiration under the modules, ET_0 , is recomputed with the FAO-56 Penman-Monteith equation (Allen et al., 1998)

from the reduced radiation and canopy temperature. Soil water conservation under the modules (Hassanpour Adeg et al., 2018; Elamri et al., 2018) therefore emerges from the model, but rain redistribution by the modules (Marrou et al., 2013a) is ignored.

Crop growth. Crop growth follows the SIMPLE model (Zhao et al., 2019) with the species parameters of its Table 1a (Table 1). Daily biomass growth is

$$\Delta B = R_{\text{crop}} \cdot \text{RUE} \cdot f_{\text{Solar}} \cdot f_{\text{CO}_2} \cdot f_{\text{Temp}} \cdot \min\{f_{\text{Heat}}, f_{\text{Water}}\}, \quad (2)$$

where $R_{\text{crop}} = sR$ is the radiation received by the crop. The terms are as follows.

- f_{Solar} is the canopy’s radiation interception. In SIMPLE it depends only on thermal time, with senescence accelerated by heat and drought stress. Shade therefore reduces the daily biomass increment but not leaf-area development.
- f_{Heat} decreases linearly between the heat threshold T_{heat} and the extreme temperature T_{ext} of the canopy maximum temperature.
- $f_{\text{Water}} = 1 - S_{\text{water}} \text{ARID}$, where the agricultural reference index for drought (Woli et al., 2012) is computed from a daily soil-water balance with curve-number runoff, drainage and root water uptake. Irrigated fields have $f_{\text{Water}} = 1$.

SIMPLE has a fixed harvest index and therefore ignores the high sensitivity of grain number to shading around flowering (Fischer, 1985; Andrade et al., 1999). It also ignores the differences in shade tolerance between crop types. In the meta-analysis of Laub et al. (2022), maize and grain legumes lost yield strongly even at low shade levels, whereas C3 cereals and tuber crops initially lost less than proportionally; shading trials also show marked effects of shade on maize development (Ramos-Fuentes et al., 2023). We added a stage-dependent penalty. During a critical window ($0.45 \leq \text{TT}/T_{\text{sum}} \leq 0.65$), the harvest index is multiplied by $1 - h\bar{d}$, where $\bar{d}$ is the mean relative radiation deficit $1 - s$ in that window. The sensitivity h is highest for maize (0.6) and soybean (0.5), the crop types that Laub et al. (2022) found most shade-sensitive, lower for the C3 cereals (wheat 0.4; rice 0.5, for which no specific curve was available) and lowest for potato (0.2).

256 These values are assumptions that reproduce this ranking; they are not cal-
 257 ibrated estimates. Section 3.6 shows the resulting shade-response curves.
 258 Because RUE is linear, SIMPLE does not credit the higher light-use effi-
 259 ciency of shaded canopies under more diffuse light. The penalty, conversely,
 260 makes shading around flowering more damaging than proportional. The net
 261 effect is closer to the field evidence than either assumption alone.

262 The thermal-time requirement to maturity was calibrated for every site as
 263 the mean thermal time between the GGCM sowing and maturity dates under
 264 the 2001–2020 climate, and the canopy parameters were scaled accordingly.
 265 The cultivar is thus adapted to the present climate, and warming shortens the
 266 season. Growth stops smoothly once the thermal-time requirement is met.
 267 Yield is biomass multiplied by the (shade-adjusted) harvest index. Each site
 268 is represented by a rainfed and an irrigated sub-field in proportion to its
 269 SPAM irrigated fraction. Each sub-field carries its own array, so both yield
 270 *and* electricity are area-weighted over the two sub-fields.

271 2.3. SALS: simulation-trained adaptive light sharing

272 SALS learns a *control network* $u_t = \pi_\omega(\mathbf{x}_t; g)$ (Algorithm 1). It is a
 273 multilayer perceptron with two hidden layers of 64 units and a bounded-logit
 274 output, $u = \sigma(5 \tanh(z/5)) \in [0.007, 0.993]$, which keeps gradients non-zero
 275 at the extremes. It chooses the daily light-sharing level of Eq. (1). Its inputs
 276 are the information available on day t :

- 277 • the crop and soil state: relative thermal time TT/T_{sum} , radiation in-
 278 terception f_{Solar} , relative soil water and the previous day’s ARID;
- 279 • a *forecast* of the day’s maximum temperature (relative to T_{heat} and
 280 T_{opt}), irradiance and ET_0 ;
- 281 • static information: the irrigation status of the sub-field, the array’s
 282 GCR g , latitude, day of season and the crop.

283 Forecasts are perturbed with random errors, independently every day: Gaus-
 284 sian with a standard deviation of 2°C for maximum temperature, and mul-
 285 tiplicative with standard deviations of 20% for irradiance and 15% for ET_0 .
 286 The physics always uses the true weather. The same forecast errors are
 287 applied to the rule-based controllers. The controller is conditioned on g .
 288 During training, each season is simulated with a GCR close to the density
 289 that is sustainable at its training site. This is the largest value on a coarse

290 grid $(0.05, 0.10, \dots, 0.50)$ at which seasonal sharing keeps the mean retention
 291 of the site’s training seasons at or above ρ , plus a uniform perturbation in
 292 $[-0.05, +0.10]$ truncated to $[0.05, 0.50]$. The network therefore learns to con-
 293 trol the densities that design selection will later pick. Outside the cropping
 294 season the trackers backtrack for maximum electricity.

295 For site-season i with open-field yield Y_i^{open} , let $r_i = Y_i/Y_i^{\text{open}}$ be the
 296 yield retention and $e_i = E_i/E_{\text{PV},i}$ the electricity relative to a conventional
 297 PV plant. Training minimises the penalised objective

$$\mathcal{L}(\omega) = \mathbb{E}_{i \sim \mathcal{B}} \left[-e_i + \mu_1 \max\{0, \rho - r_i\} + \mu_2 \max\{0, \rho - r_i\}^2 \right], \quad \rho = 0.9, \mu_1 = 5, \mu_2 = 30, \quad (3)$$

298 which combines an exact (hinge) and a quadratic penalty for the food-security
 299 floor ρ . Because r_i and e_i are outputs of the differentiable simulator, the
 300 exact gradient of Eq. (3) is obtained by back-propagation through the 365
 301 daily steps of every season (back-propagation through time). This is the key
 302 difference from reinforcement learning with black-box simulators (Gautron
 303 et al., 2022), which estimates gradients from sampled returns.

304 Each minibatch contains 512 site-seasons, each simulated as a rainfed and
 305 an irrigated sub-field, drawn from one of the four climates in turn. Seasons
 306 with a failed open-field crop ($Y^{\text{open}} < 0.2 \text{ t ha}^{-1}$) are excluded from training
 307 and evaluation. The network is fine-tuned through the simulator with Adam
 308 (Kingma and Ba, 2015) (learning rate 10^{-3} , cosine decay, gradient-norm
 309 clipping at 1) for 300 steps. Growth-stage logistic functions are evaluated in
 310 a numerically stable form, so that gradients remain finite even under extreme
 311 heat.

312 *Initialisation.* Trained from random weights, the controller converged to
 313 sharing light throughout the cropping window ($u > 0.99$ on all days), i.e. to
 314 the seasonal-sharing rule, because the gradient of electricity with respect to u
 315 is small and of the same sign for most days of a season. We therefore initialise
 316 the network by behaviour cloning of the phenology rule of Section 2.4: binary
 317 cross-entropy against the rule’s output (soft targets 0.95 and 0.05) on about
 318 1.7 million day-states visited by the rule in the training seasons, with 1500
 319 Adam steps at a learning rate of 3×10^{-3} . Fine-tuning then moves the
 320 policy away from this initialisation wherever doing so improves the objective.
 321 Consequently, the gain of SALS over the phenology rule measures what fine-
 322 tuning through the simulator adds to that rule, not what can be learned
 323 without it.

Algorithm 1 SALS training

Require: training site-seasons in four climates; floor ρ ; penalties μ_1, μ_2

- 1: initialise π_ω by behaviour cloning of the phenology rule
- 2: **for** step = $1, \dots, N$ **do**
- 3: sample a climate and a minibatch $\mathcal{B}$ of site-seasons (rainfed and irrigated sub-fields)
- 4: set g_i to the site’s sustainable density plus a perturbation; initialise crop and soil states
- 5: **for** day $t = 1, \dots, 365$ **do**
- 6: $u_{it} \leftarrow \pi_\omega(\mathbf{x}_{it}; g_i)$ from state and noisy forecast if the crop is in the field, else 0
- 7: hourly rotation (Eq. 1), PV electricity, crop light, canopy temperature, ET_0
- 8: advance the soil-water balance and the SIMPLE crop state (Eq. 2)
- 9: $r_i \leftarrow Y_i/Y_i^{\text{open}}, e_i \leftarrow E_i/E_{\text{PV},i}$ (area-weighted); loss $\mathcal{L}$ (Eq. 3)
- 10: back-propagate through the season; Adam update of ω

324 *2.4. Deployable design selection and baselines*

325 A real array has one fixed density, which must be chosen before it is built.
326 Every method, SALS included, therefore uses the same *history-based design*
327 *rule*. For a site, the GCR is the largest value on the grid $\{0.05, 0.075, \dots, 0.50\}$
328 for which the method’s mean yield retention over the site’s 2001–2014 seasons
329 is at least $\rho = 0.9$. If no value qualifies, no array is built and the site
330 remains open field. A stricter *chance-constrained* variant requires $r \geq 0.9$ in
331 at least 80% of the historical seasons. The historical seasons of the test sites
332 are never used for training, and the weather of the evaluation years is never
333 used for design. We compare four controllers:

- 334 • *AV static*: backtracking trackers without light sharing ($u = 0$);
- 335 • *Seasonal sharing*: the light-sharing rotation ($u = 1$) throughout the
336 cropping season;
- 337 • *Stress rule*: an expert rule that shares light ($u = 1$) during the main
338 growth phase ($0.1 < TT/T_{\text{sum}} < 0.9$) except on days with a heat-stress
339 forecast ($T_{\text{max}} \geq T_{\text{heat}}$) or drought ($\text{ARID} > 0.5$), when the modules
340 track the sun and shade the crop;

341 • *SALS*: the learned controller.

342 The open field and a conventional PV plant without crops (LER = 1 each)
 343 are the references. Because the design rule is identical for all controllers, dif-
 344 ferences between methods reflect the quality of control and the array density
 345 that each controller can sustain under the food-security floor. We also com-
 346 pare the controllers at identical GCR over the whole grid (matched-density
 347 frontiers).

348 2.5. *A shade-response model learned from field data*

349 The simulator needs a crop response to shade that is supported by ob-
 350 servations. We learned one from the open data of the meta-analysis of Laub
 351 et al. (2022) (Zenodo 10.5281/zenodo.5716091, CC BY 4.0): 428 observa-
 352 tions of yield relative to an unshaded control at relative shade reductions of
 353 0–90% from 58 studies, with crop and crop type, experiment type (artificial
 354 shading or intercropping), climate zone, irradiation and shade type. Rela-
 355 tive yield is modelled as $y(s) = \exp(-\beta s^\gamma)$ with Student- t errors ($\nu = 4$;
 356 maximum a posteriori estimation, so this is a hierarchical shrinkage regres-
 357 sion with bootstrap intervals and not a full Bayesian model). We compared
 358 six models: proportional loss $y = 1 - s$; one curve for all crops; one curve
 359 per crop type without pooling; one curve per crop type with partial pooling
 360 (Gaussian shrinkage of $\log \beta$ and $\log \gamma$ towards a common value, scale 0.5);
 361 partial pooling with covariates on $\log \beta$ (standardised irradiation, subtrop-
 362 ical zone, solar-panel shade and intercropping); and gradient-boosted trees
 363 on the same inputs. Fits use L-BFGS. Models were compared by leave-one-
 364 study-out cross-validation, in which every study is held out once, which tests
 365 the prediction of a new study and not of a new observation in a known study.
 366 Intervals for the crop-type curves and covariate effects come from a cluster
 367 bootstrap over studies (150 resamples).

368 2.6. *Shade-response uncertainty: gradient calibration, robust controllers and* 369 *design*

370 The crop’s response to shade is the least certain part of the simulator,
 371 so we treat it as an unknown with a distribution. Two parameters per crop
 372 enter the simulator: a radiation-use compensation c (growth is multiplied by
 373 $1 + c(1 - s)$, s being the share of open-field irradiance that reaches the crop)
 374 and a multiplier h on the flowering-window harvest-index penalty. Because
 375 the simulator is differentiable, both can be fitted by gradient descent through

376 whole simulated seasons to published yield–shading observations. The ob-
 377 servations are the relative yields at 20% and 40% shading of the data-driven
 378 curves of Section 2.5 for the five crops (median, with an error equal to the
 379 width of the 80% bootstrap interval divided by 2.56 and at least 0.04; rice
 380 takes the curve of C3 cereals, which contains no rice), the field trials on rice,
 381 wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-
 382 Fuentes et al., 2023) (each season of each trial as a separate observation, error
 383 0.06–0.08) and the unaffected harvest index of shaded maize (Ramos-Fuentes
 384 et al., 2023). We call the resulting ensemble E2. A wider ensemble, E3, uses
 385 the same procedure with a target error of 0.2 for the learned curves, which is
 386 the between-study spread of a new study (Section 3.7). An earlier ensemble,
 387 E1, was fitted in the same way to curves read by hand from the figures of
 388 Laub et al. (2022) and was used to train the robust and adaptive controllers
 389 below. Priors are $c \sim \mathcal{N}(0.4, 0.5^2)$ and $h \sim \mathcal{N}(1, 0.5^2)$, and the param-
 390 eters are clamped to $[0, 2.5]$ (c) and $[0, 2]$ (h) after every step, which is why
 391 a response milder than the data cannot be lowered further. We fitted 12
 392 ensemble members by randomised maximum a posteriori estimation: every
 393 member sees the observations perturbed by their errors and a prior draw
 394 perturbed by its width, and is optimised with Adam (40 steps, with the loss
 395 still falling slightly at the last step; 40 training-block seasons per crop and
 396 member; shading of the static arrays cached). The ensemble is a device for
 397 generating plausible responses and not a validated posterior: it has no cov-
 398 erage check, and for some parameters (the harvest-index multiplier of wheat
 399 and potato) its spread equals that of the prior. The design analyses use five
 400 fixed members (indices 0, 3, 5, 8 and 10 of 12), chosen in advance. The
 401 ensemble is used in three ways. All training and evaluation ran on a 4-core
 402 CPU; one training run takes about 20–25 minutes and the calibration of one
 403 ensemble about 7 minutes.

404 *Robust and adaptive controllers.* SALS-robust has the architecture and
 405 training of SALS (Section 2.3), but every training season is simulated with the
 406 response of a randomly drawn ensemble member that the controller does not
 407 observe, and the density jitter is widened to $[-0.05, +0.25]$ because feasible
 408 densities differ between members. SALS-adaptive additionally observes the
 409 realised biomass of the crop under the modules relative to an open reference
 410 plot, a quantity that can be measured on the farm by remote sensing or by
 411 a reference plot. A feedback rule that shares light whenever this ratio falls
 412 below a target serves as the hand-written counterpart; its target (1.0) was
 413 chosen on training-block seasons under the ensemble.

414 *Uncertainty-aware design.* The design rule of Section 2.4 chooses the
 415 largest GCR whose historical retention meets the floor. Under shade-response
 416 uncertainty we choose the largest GCR that meets the floor in at least 75%
 417 of five ensemble members, and compare it with a design under the ensemble
 418 mean and with an oracle design under the true response. Evaluation of this
 419 rule is leave-one-out: the truth is the held-out member and the design uses
 420 the other four.

421 *Stress tests.* Controllers and rules are also evaluated under four fixed
 422 responses, the proportional response, the response calibrated to the meta-
 423 analysis, the response fitted to the field trials and a harsh response (propor-
 424 tional growth and a 1.5 times larger harvest-index penalty). The proportional
 425 and harsh responses lie outside the range of the ensemble.

426 2.7. Economic analysis

427 To judge whether the electricity gains matter economically, we computed
 428 an annual net value per hectare for each method relative to the open field:
 429 electricity revenue minus the annuity and operating cost of the installed ca-
 430 pacity minus the loss of crop revenue (crop production costs are unchanged).
 431 The installed capacity is 2.1 g MWp per hectare (modules cover a share g of
 432 the land at 0.21 kWp m^{-2}). Yield is dry matter, valued at assumed prices.
 433 A conventional ground-mounted plant (GCR 0.4) that replaces the crop is
 434 the comparator. All parameters (Table 14) are assumptions chosen to be
 435 representative, not data, and their ranges drive the conclusions; we therefore
 436 report break-even electricity prices and one-at-a-time sensitivities, and we do
 437 not monetise food security, which market crop prices do not capture.

438 2.8. Evaluation protocol, statistics and upscaling

439 Sites were split spatially: $10^\circ \times 10^\circ$ blocks were randomly assigned to train-
 440 ing (70%) or test (30%), so that no test site has a training site in the same
 441 block. Training uses the 2001–2014 seasons of the training sites. Design
 442 selection uses the 2001–2014 seasons of the test sites. Evaluation uses the
 443 2015–2019 seasons of the test sites. The same site-seasons are analysed in all
 444 four climates.

445 We report the following metrics:

- 446 • the mean electricity ($\text{MWh ha}^{-1} \text{ yr}^{-1}$) and the relative electricity e ;
- 447 • the mean yield retention r and the production-weighted retention $\sum Y / \sum Y^{\text{open}}$;

Table 1: SIMPLE crop parameters (Zhao et al., 2019, Table 1a) and the harvest-index shade sensitivity h added in this study. T_{sum} , I_{50A} and I_{50B} are rescaled per site to the local crop calendar (Section 2.2). Units: temperatures in °C, thermal times in °C d, RUE in gMJ⁻¹; S_{CO_2} is the relative RUE increase per 100 ppm above 350 ppm.

Crop (cultivar)	h	T_{sum}	HI	I_{50A}	I_{50B}	T_{base}	T_{opt}	RUE	$I_{50 \text{ max } H}$	$I_{50 \text{ max } W}$	T_{heat}	T_{ext}	S_{CO_2}	S_{water}
Wheat (Batten)	0.4	2150	0.34	280	50	0	15	1.24	100	25	34	45	0.08	0.4
Rice (IR72)	0.5	2300	0.47	850	200	9	26	1.24	100	10	34	50	0.08	1.0
Maize (McCurdy 84aa)	0.6	2050	0.50	500	50	8	28	2.10	100	12	34	50	0.01	1.2
Soybean (Williams 82)	0.5	2350	0.40	600	200	6	27	0.86	120	20	36	50	0.07	0.9
Potato (Sebago)	0.2	2400	0.85	500	350	4	22	1.30	50	30	34	45	0.10	0.4

- season-level compliance (share of seasons with $r \geq \rho$) and site-level compliance (share of sites whose mean test-season retention is $\geq \rho$);
- the lower tail (10th percentile of r , share of seasons with $r < 0.8$);
- the land-equivalent ratio $\text{LER} = r + e$ (Dupraz et al., 2011).

Statistical comparisons use site means as the unit of analysis. We report paired Wilcoxon signed-rank tests and cluster (site) bootstrap 95% confidence intervals of the mean difference. SALS was trained with three random seeds on the main split, and with one seed on each of two further spatial splits.

For the global estimate, every site is evaluated in the split in which it is held out. The mean electricity and yield per hectare of each crop are scaled to the *physical* area of that crop in SPAM 2010, which avoids double counting multiple-cropped land (Waha et al., 2020). We report the installed capacity (TWp), the capacity factor and a supply curve.

3. Results

3.1. Electricity and food on held-out sites and years

Table 2 compares the four controllers on the 92 held-out sites of the main split (seasons sown in 2015–2019), after each site’s density has been chosen from its 2001–2014 seasons with the same rule for all controllers. By construction all methods keep the mean yield retention close to 90% (0.904–0.917), so differences in electricity are differences in the density that each controller can sustain under this floor.

Static arrays are the weakest. Without light sharing, the floor limits the ground coverage to 6% on average, and arrays produce 217 MWh ha⁻¹ yr⁻¹,

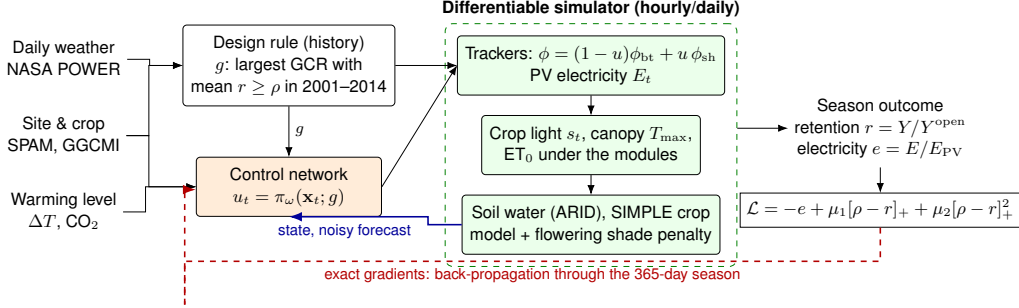


Figure 1: Simulator and controller. The array density g of a site is fixed from its historical seasons. A control network maps the daily crop and soil state, the density g and a noisy forecast to the light-sharing level u_t . The differentiable simulator converts these decisions into PV electricity, light and microclimate at the crop, soil water and crop growth. The season’s electricity and yield retention enter a penalised objective whose exact gradient is back-propagated through the 365-day season.

472 16% of a conventional PV plant per hectare (LER 1.07). Rotating the
 473 modules to let light through during the cropping season (seasonal shar-
 474 ing) allows a coverage of 17% and 397 MWh ha⁻¹ yr⁻¹ (LER 1.19). A rule
 475 that shares light only while the crop is growing (phenology rule) reaches
 476 465 MWh ha⁻¹ yr⁻¹ (LER 1.24). An expert rule that additionally shades the
 477 crop on forecast heat or drought days (stress rule) is worse than the phenol-
 478 ogy rule (LER 1.18): its density is limited to 13% because it reduces light at
 479 times when the floor is most binding.

480 **The learned controller is best by a modest, consistent margin.**
 481 SALS sustains a density of 17% (similar to the phenology rule), produces
 482 510 MWh ha⁻¹ yr⁻¹ and reaches an LER of 1.270 at the same retention. Com-
 483 pared with the phenology rule this is +0.038 in LER (95% CI over sites
 484 0.025–0.053; +10% electricity) and it is higher at 84% of the 92 sites (Ta-
 485 ble A.17). Compared with static arrays it is +0.205 (0.176–0.236), with 2.35
 486 times the electricity at 2.9 times the density. The frontier in Fig. 3 shows
 487 the same picture at matched densities: the learned policy lies on or slightly
 488 above the phenology rule over the whole density range, both lie well above
 489 seasonal sharing and the stress rule, and static arrays are far below.

490 **What the floor does and does not guarantee.** Enforcing the floor
 491 on the historical mean does not guarantee it in every season. In the test
 492 seasons, 70% of SALS seasons have $r \geq 0.9$ (phenology rule 69%, stress rule
 493 76%, static arrays 93%), and 2.7% have $r < 0.8$ (1.3%, 0.9% and 0.9%).

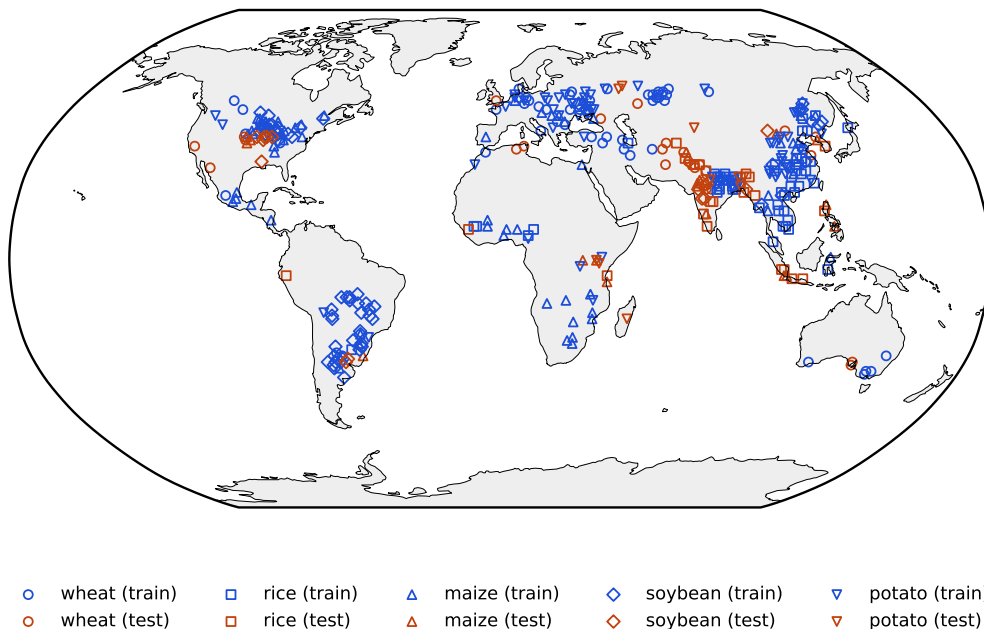


Figure 2: The 400 crop–site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites of the main split.

494 The lower tail of SALS is therefore slightly heavier than that of the rules:
495 its 10th percentile of r is 0.883, against 0.887, 0.886 and 0.903. Requiring
496 $r \geq 0.9$ in at least 80% of a site’s historical seasons (chance-constrained
497 design) raises the season-level compliance to 90–94% for the light-sharing
498 controllers. It costs about 0.065 in LER for SALS (1.205), which remains the
499 best controller (phenology rule 1.192; Section 3.2).

500 3.2. Robustness

501 **Seeds and splits.** Three training seeds of SALS give LERs of 1.270,
502 1.268 and 1.269 under the baseline climate and 1.297, 1.298 and 1.297 at
503 $+3^\circ\text{C}$. On two further spatial splits, in which different blocks of sites are
504 held out, SALS exceeds the phenology rule by 0.047 and 0.041 in LER under
505 the baseline climate and by 0.052 and 0.039 at $+3^\circ\text{C}$ (main split: 0.034 and
506 0.032). In every split SALS ranks first, the phenology rule second and static
507 arrays last.

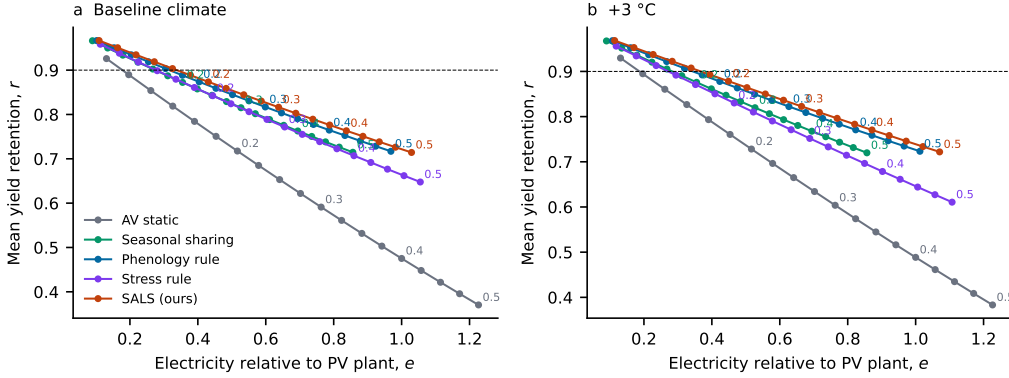


Figure 3: Electricity-retention frontiers at identical array densities (labels: GCR) for the held-out sites. Each point is the mean over test seasons at a fixed GCR; the dashed line is the 90% floor.

508 **Canopy cooling.** The canopy-cooling coefficient κ is the most un-
 509 certain microclimate parameter. We repeated the whole protocol, includ-
 510 ing the historical design step, with $\kappa = 0, 1$ and 2°C . Electricity changes
 511 with κ because cooler canopies tolerate denser arrays (SALS: 475, 510 and
 512 $549 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ under the baseline climate), but the retention is held at
 513 0.905 by the design rule and the ranking is unchanged. SALS exceeds the
 514 phenology rule by 0.029, 0.034 and 0.042 in LER under the baseline climate
 515 (0.022, 0.032 and 0.051 at $+3^\circ\text{C}$).

516 **Which ingredients matter.** Two ablations test the design of SALS
 517 at the same training budget. Masking the stress inputs (soil water, ARID,
 518 and the maximum temperature, irradiance and ET_0 forecasts) leaves the
 519 LER unchanged (1.269 vs 1.270 under the baseline climate, 1.297 vs 1.297
 520 at $+3^\circ\text{C}$). Training only on the baseline climate gives 1.268 and 1.291. The
 521 improvement over the rules therefore does not depend on stress-aware inputs
 522 or on training in warmed climates.

523 3.3. Floor conditioning, chance-constrained training and forecast errors

524 **One network for all floors.** Regulations and farmers set different
 525 floors. The network of the main experiments was trained at $\rho = 0.9$ only.
 526 We evaluated it at floors of 0.80, 0.85, 0.90 and 0.95, re-running the historical
 527 design step for each floor (Table 3). Its LER exceeds that of the phenology
 528 rule by 0.037, 0.036, 0.034 and 0.033 under the baseline climate (0.040, 0.037,
 529 0.032 and 0.033 at $+3^\circ\text{C}$), so the benefit carries over to other floors without

retraining. A network that received the floor as an additional input and was trained over all four floors did not improve on it (differences in LER ≤ 0.001). Because the design step already adapts the density to the floor, conditioning the controller on the floor is not needed.

Chance-constrained training. We also trained a network whose objective penalises a batch share of seasons below the floor that exceeds 20%. It did not improve the tail. With the mean-based design step its LER was 1.263 (baseline) and 1.290 (+3 °C) against 1.270 and 1.297, and with chance-constrained design it reached season-level compliance of 92.6% against 92.0% and an LER of 1.198 against 1.205. The season-level guarantee is obtained by the design step (Section 3.1), not by the training objective.

Forecast errors. We raised the forecast errors seen by the controllers to 0, 1, 2 and 3 times the errors used in training. The LER of SALS and of the phenology rule was identical at all four error levels to three decimals (densities 0.2 and 0.3: SALS 1.307 and 1.457, phenology rule 1.278 and 1.416). Neither controller uses the weather forecast: both rely on the crop’s thermal time, which is observable from on-farm temperature records. The stress rule, which depends on the forecast of maximum temperature, degrades with forecast error: at density 0.3 its retention falls from 0.778 to 0.751 and its LER from 1.432 to 1.418 as the errors triple.

A tuned rule. The phenology rule above was written, not tuned, whereas SALS is trained. We therefore tuned the window of the rule on the training-block seasons with the same objective (electricity at the density where mean retention equals the floor); the best window starts at 0.15 instead of 0.05 of the thermal-time requirement (Table 4). Pooled over three splits and 76 distinct crop–site pairs, the tuned rule raises LER over the original rule by 0.026 (95% cluster-bootstrap interval 0.016 to 0.037), and SALS exceeds the tuned rule by 0.018 (0.012 to 0.024), or 5.7% more electricity at equal retention, and the original rule by 0.044 (0.030 to 0.060). Tuning the rule thus accounts for about 60% of the margin of SALS over the written rule. A five-parameter ramp-plateau schedule optimised by CMA-ES on the same training seasons performs like the tuned window (LER 1.238 against 1.233, difference +0.005, interval -0.001 to $+0.010$), and SALS exceeds it by 0.013 (0.007 to 0.019); the optimum is an almost binary window (start 0.17, end 0.93, level 0.99), so more flexible schedules of this family do not close the gap. The remaining margin of SALS is small but not zero; it is not explained by the tuning of a window or by a smooth ramp, and we did not test richer rule families (for example schedules that depend on soil water).

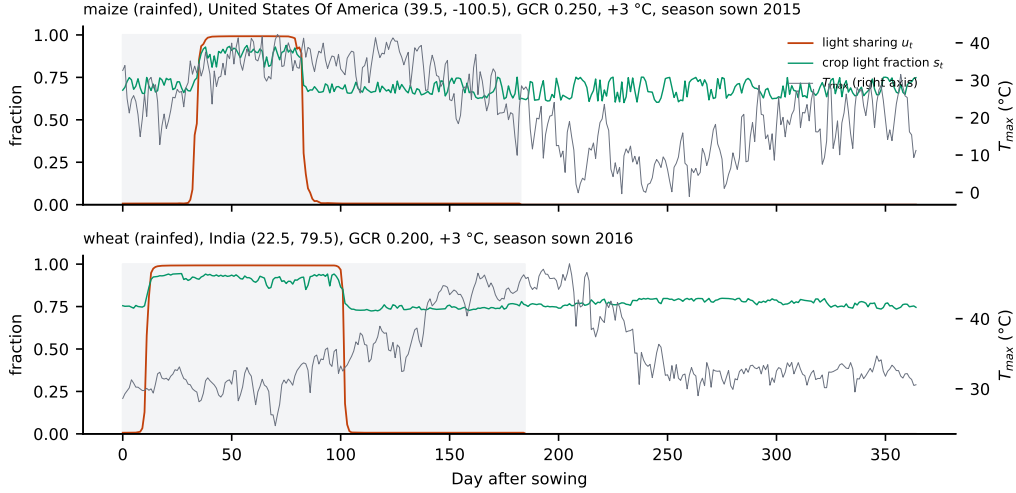


Figure 4: Learned daily control for two test seasons under $+3^{\circ}\text{C}$, at the density chosen for the site, simulated for the dominant sub-field of the site. Shading marks the cropping window. Orange: light-sharing level u_t ; green: fraction of open-field irradiance reaching the crop; grey: daily maximum temperature (right axis).

568 SALS meets the floor in fewer seasons than the rules (83% against 84–85%).

569 3.4. What the learned policy does

570 Averaged over the rainfed test seasons of the main split at a density of 0.3,
 571 the light-sharing level of SALS is a smooth function of relative thermal time.
 572 It is close to zero in the first 10% of the thermal time required for maturity,
 573 rises through 0.55 at 10–20%, stays at 0.94–0.99 between 30% and 70%, and
 574 falls through 0.58 at 80–90% to about zero at maturity. The plateau covers
 575 the window in which our crop model penalises shading through the harvest
 576 index. The ramps at both ends let the modules produce electricity when
 577 additional light is worth least to the crop. Fig. 4 shows two example seasons.
 578 The phenology rule, which is also the initialisation of the network, shares
 579 light during the whole growing period. Fine-tuning through the simulator
 580 found that sharing can be reduced at the edges of the season: SALS gains its
 581 advantage by yielding light when it matters and collecting electricity when
 582 it matters less.

583 The policy also shares less light on days with a forecast heat stress: the
 584 mean level is 0.35 on such days against 0.60 (baseline) and 0.51 ($+3^{\circ}\text{C}$)
 585 on other days, and 58–63% of sites show a lower level on heat days. This

cannot be read as a stress response, however. Heat days occur late in the season, when the phenological ramp is already falling, and a policy without any stress input shows a similar contrast (0.37 vs 0.57 in a separate check). The ablation above confirms that stress information does not improve the result.

3.5. Crops and regions

The benefit of light sharing differs between crops (baseline climate, SALS). Soybean and maize tolerate the densest arrays (GCR 0.24 and 0.19; 691 and 577 MWh ha⁻¹ yr⁻¹; LER 1.41 and 1.30), potato follows (0.19; 542; 1.31), and rice and wheat are the most restrictive (0.13 and 0.15; 395 and 428; 1.20 for both). The stress rule has a lower LER for every crop (1.26 for soybean, 1.14–1.21 for the others).

The floor is not met evenly. Production-weighted retention is 0.888 for soybean and 0.898 for maize, below 0.9, because high-yielding seasons lose more than low-yielding ones. Regionally, Latin America and Oceania (4 sites) combine the highest density (GCR 0.41) and electricity (1090 MWh ha⁻¹ yr⁻¹) with a production-weighted retention of only 0.862. Africa and the Middle East (8 sites) reach 0.894, and only 63% of their sites keep a mean retention of at least 0.9. South Asia, the region with the most sites (39), reaches 0.902 (67% of sites) at a density of 0.14. A regulatory floor enforced per site would therefore need a region- and crop-specific design step, not a global one.

3.6. Model behaviour and validation

PV output. The simulated specific yield of an isolated tracker row agrees with PVGIS 5.2 (European Commission, Joint Research Centre, 2022) at 29 sites with $r = 0.971$, a mean bias of -4.9% and a mean absolute deviation of 5.5% (Section 2.2).

Shade response. Fig. 5 shows the simulated relative yield of rainfed crops under static arrays as a function of the season-mean shading of the crop. Yield declines close to proportionally with shading: 84–89% relative yield at 12% shading, 70–78% at 25%, 57–67% at 37% and 35–46% at 57%. Wheat loses most and potato and maize least. This response is steeper than the *less than proportional* losses that Laub et al. (2022) report for small shade levels of C3 cereals and tubers, and it does not reproduce their finding that maize and grain legumes are more shade sensitive than C3 cereals. In the simulator, shading also relieves heat and drought in rainfed maize and soybean, which

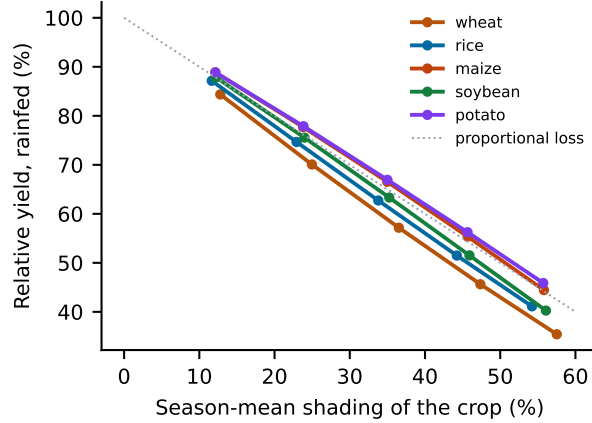


Figure 5: Simulated relative yield of rainfed crops under static arrays (GCR 0.1–0.5) against season-mean shading of the crop, baseline climate, held-out sites. The dotted line is proportional loss.

offsets the higher harvest-index penalty we assigned to them. The absolute effect of shading on each crop is therefore uncertain (Section 4.5).

Comparison with field trials. We fitted one radiation-use compensation coefficient per crop to the yield–shading curves of Laub et al. (2022) using training-block sites only (Methods, calibration script in the repository). The fit changed wheat and rice only (coefficients 0.4 and 0.1) because the proportional model already lies within about 0.02 of the C3-cereal and tuber curves, and it cannot lower the maize and legume responses below the proportional case (root-mean-square error 0.16 for maize and 0.08 for soybean). Against independent field trials (Table 5) the model underestimates retained yield: rice 0.79 against 0.84 (Gonocruz et al., 2021), wheat and potato 0.74 and 0.72 against 0.81–0.82 in the worse season of Weselek et al. (2021) (and above 1 in the better one), and maize 0.70–0.75 against 0.83–0.91 at 30–35% shading (Ramos-Fuentes et al., 2023), where harvest index was not affected by shade. The model is therefore conservative for food retention with respect to these trials, except for the stronger maize losses of the meta-analysis. Table 6 shows the consequence for the design results: with the calibrated response the allowed density and the LER of every method rise (phenology rule 1.28 against 1.23), and with a response fitted to the field trials the LER of the phenology rule is 1.62. The phenology rule is the best method in all three cases, and light sharing adds LER +0.17, +0.19 and

+0.30 over static arrays (conservative, calibrated, field-fitted); only the order of the seasonal and stress rules changes. The absolute gains for rice, wheat and potato are thus conservative with respect to these trials; for maize and soybean the meta-analysis points the other way (Section 3.7).

3.7. *A shade-response model learned from field data*

Cross-validation by study shows what the data can and cannot support (Tables 7 and 8). Over all 339 shaded observations the root-mean-square error of a held-out study falls from 0.43 for proportional loss $y = 1 - s$ to 0.37 for one curve for all crops and to 0.31 for crop-type curves, but this gain comes from crop types that the simulator does not represent (berries, forages, fruits and vegetables: 0.53 to 0.36). For the four crop types that the simulator uses, the learned curves are not better than proportional loss: the error is 0.222 against 0.220 pooled (difference +0.002, 95% interval -0.023 to $+0.030$), 0.187 against 0.214 for C3 cereals, and slightly worse for maize, grain legumes and tubers. Partial pooling is as good as separate curves (0.312 against 0.313) and the covariates add almost nothing (0.311). Gradient-boosted trees on the same inputs are worse (0.321) than the parametric model. The error of 0.22 for the four crop types is large: yields of a new study differ from the learned curve by about 20 percentage points, which is the between-study spread that a design under uncertainty has to cover. Crop type is the information that transfers between studies, and none of the covariate effects on $\log \beta$ is clearly different from zero (irradiation -0.10 , 90% interval -0.32 to 0.09 ; subtropical zone -0.37 , -0.73 to 0.07 ; intercropping 0.16 , -0.48 to 0.45). The estimated effect of solar-panel shade is not interpretable, because all 46 panel observations are of leafy vegetables, mainly lettuce, and the effect is confounded with crop type.

The learned curves show a clear ordering of crop types (Table 9). At 20% shading C3 cereals and tubers lose less than proportionally (relative yield 0.93), whereas maize (0.73) and grain legumes (0.71) lose more; at 40% shading the values are 0.72 and 0.71 against 0.42 and 0.40. Intervals are wide for maize (0.27–0.59 at 40%) and for tubers, which rest on two studies. The simulator’s responses are close to the data for cereals and potato but, under both calibrated ensembles, too mild for maize and soybean at 40% shading (0.75 and 0.48 for E2 against 0.42 and 0.40 in the data). The calibration cannot lower the maize response further, because the simulated shading also relieves heat and drought stress and the compensation and harvest-index

parameters are bounded below at zero. The field trials of maize (Ramos-Fuentes et al., 2023) are also milder than the meta-analysis. For maize and soybean the direction of the bias is uncertain, and the gains reported in this paper are possibly overestimated.

3.8. Learning under shade-response uncertainty

Calibration. Gradient descent through the simulator fitted the five crops to the data-driven curves and the field trials (ensemble E2, Table 10). The data identify two features: the harvest index of maize is hardly affected by shade (multiplier 0.16 ± 0.13 , $c \approx 0.14$), as observed by Ramos-Fuentes et al. (2023), and soybean is more shade sensitive than the cereals (compensation 0.05, harvest-index multiplier 1.66), as in Laub et al. (2022). Wheat has the largest compensation (0.91 ± 0.31) and the largest uncertainty; the compensation is an effective parameter that absorbs diffuse-light and light-saturation effects that the simulator does not model explicitly, not a physical value.

Learned controllers. Controllers were evaluated under four fixed responses on 21 crop–site pairs of split 0 with one training seed, so the intervals below, which are cluster bootstraps over pairs of per-pair mean LER differences, do not include seed or split variation (Table 11 gives season means, which differ slightly). Under every response SALS, trained on the proportional response, is better than the written phenology rule: by +0.019 (0.007 to 0.039), +0.024 (0.009 to 0.045), +0.038 (0.017 to 0.063) and +0.015 (0.000 to 0.036) LER under the proportional, calibrated, field-fitted and harsh responses. Training on the narrow ensemble E1 did not change this: SALS-robust differs from SALS by -0.005 to $+0.003$ (intervals contain zero except for the proportional response, 0.000 to 0.003), and SALS-adaptive, which sees the measured biomass ratio, by -0.010 to $+0.007$ (all intervals contain zero). Training on the wider ensembles did not help either: SALS-robust on E2 and E3 and SALS-adaptive on E3 are below SALS by 0.014, 0.017 and 0.017 LER under the proportional response, 0.017, 0.021 and 0.020 under the calibrated response, 0.004, 0.005 and 0.011 under the field-fitted response (intervals contain zero) and 0.008, 0.014 and 0.017 under the harsh response; the intervals exclude zero in 6 of the 9 comparisons for the proportional, calibrated and harsh responses. The E3 robust controller is not distinguishable from the written rule ($+0.001$ to $+0.003$, intervals contain zero) except under the field-fitted response ($+0.032$, 0.001 to 0.067). With one seed and 21 pairs these controllers therefore did not retain the margin of SALS within our training budget; a controller that must perform across plausible responses is

less specialised, but a larger budget or more seeds could change this. The feedback rule is worse than the written rule by 0.026–0.058 LER under every response (intervals exclude zero). A plausible reason is that the biomass ratio does not reveal the harvest-index loss from shading around flowering, but we did not test this explanation.

Design under uncertainty. The larger risk is in the design step (Tables 12 and 13, Fig. 6). The designs below are evaluated by leaving one ensemble member out as the truth and using the other four, so that the truth never enters the design. Pooled over the three spatial splits (76 distinct crop–site pairs), a density chosen under the mean response of E2 leaves 70% of the pairs at the floor when the true response is another plausible one, against 87% for an oracle design (E3: 71% and 85%). Choosing the density so that the floor is met in at least 75% of the other members raises compliance by +7.9 points under E2 (95% cluster-bootstrap interval 4.5 to 11.6) and +7.9 under E3 (3.9 to 12.4), at a cost of 0.031 and 0.025 in LER. Raising the floor used in the design from 0.90 to 0.92 under the mean response does more: +18.0 points (13.4 to 22.6) at 0.068 LER under E2 and +14.2 points (9.7 to 19.2) at 0.056 under E3, and the pairs below the floor fall short by less (mean shortfall 0.0022 against 0.0045 for the quantile rule under E2). The two kinds of rule lie on one frontier (Fig. 6): at about 88–89% compliance the ensemble rule that requires all other members to meet the floor gives LER 1.412 under E2 against 1.426 for a floor of 0.92, and under E3 1.409 against 1.458, so a simple margin is as good as or better than the ensemble rule by 0.01 to 0.03 LER. The ensemble is thus useful for sizing the margin (0.02 to 0.04 above the floor for 85–97% of the pairs), not for replacing it. None of these rules is safe outside the ensemble (Table 13): under the proportional, calibrated and harsh responses the mean-response design leaves 7–17% of the pairs at the floor, a floor of 0.92 or the ensemble rules 12–33%, a floor of 0.94 29–47% and a floor of 0.96 51–78%, at a cost of about 0.2 LER, whereas the field-fitted response, which lies inside the range spanned by the ensembles, is handled well by every rule (80–99%) except the mean-response design (68–79%). The guarantee therefore holds only within the range of responses that the ensemble spans, and a response harsher than the data supported by the ensemble breaks it.

3.9. Economics of light sharing

All figures in this section follow from the assumed parameters of Table 14 and should be read as a sensitivity analysis, not a forecast. The economics

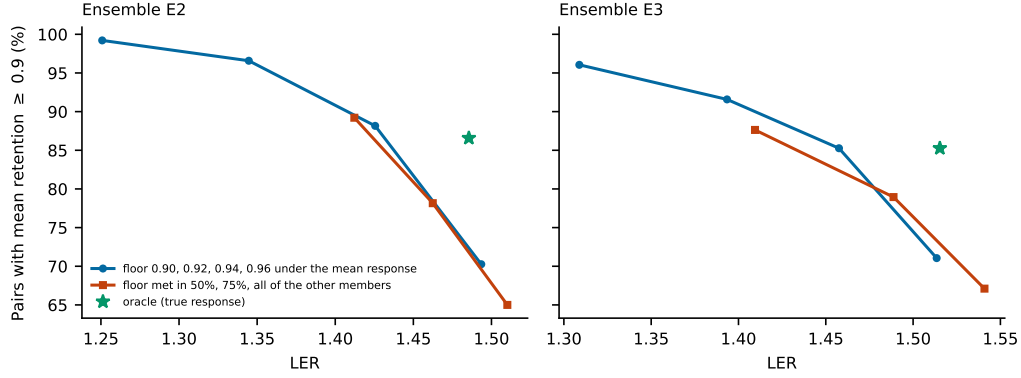


Figure 6: Share of crop-site pairs at the floor against LER for the phenology rule under design rules for shade-response uncertainty (leave-one-member-out, three splits, 76 pairs): a higher floor under the mean response (blue; the first point is the design under the mean response) and a floor met in a share of the other members (orange), with the oracle design (green).

are dominated by the capacity installed, not by the crop (Table 15): the annuity of the modules is 11–32 kUSD per hectare and year, whereas the crop revenue lost is about 0.3 kUSD, so a method is worth what it earns per installed watt. With the densities designed for the floor, static arrays earn +0.3 kUSD ha⁻¹ yr⁻¹ at 60 USD MWh⁻¹ and break even at 59, whereas all light-sharing methods lose money at that price (−3 to −10 kUSD) and break even at 70 (stress rule), 73 (SALS), 77 (tuned phenology rule), 80 (written rule) and 92 (seasonal sharing) USD MWh⁻¹. At 90 USD MWh⁻¹ static arrays (6.3 kUSD) are better than the written (4.0) and tuned (5.5) rules and worse than SALS (7.6), which overtakes static arrays above about 85 USD MWh⁻¹. A conventional plant that replaces the crop breaks even at 48 USD MWh⁻¹ and is more profitable than every agrivoltaic option above that price (16 kUSD at 60, 94 kUSD at 120, against −6 and 21 for SALS); below it all options lose money and static arrays lose least (at 30 USD MWh⁻¹: plant −23.2, SALS −18.7, static −5.7 kUSD). The break-even price of SALS exceeds that of the plant by about 25 USD MWh⁻¹, which is the order of the premium or the value of food production that agrivoltaics needs for parity with the open field; parity with the plant needs more. The sensitivity is large (Table 16): at 90 USD MWh⁻¹ the net value of SALS changes from +17 to −8 kUSD between module costs of 0.7 and 1.5 USD Wp⁻¹, whereas crop prices matter little.

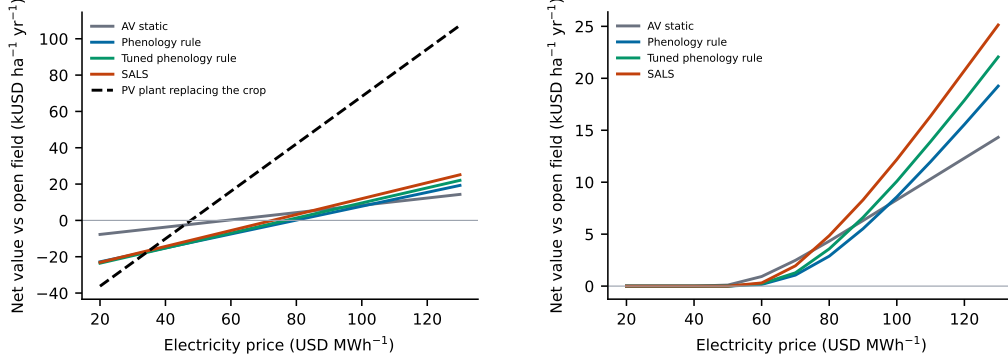


Figure 7: Net value per hectare relative to the open field against electricity price. Left: densities designed for the floor, with a conventional plant replacing the crop (dashed). Right: each site chooses the feasible density that maximises its historical net value. Illustrative parameters of Table 14.

Because the density was designed for the floor and not for profit, we also let each site choose, among the densities that meet the floor in its history, the one that maximises its historical net value (sites where none pays stay open field; Fig. 7). Then no method pays at 30–40 USD MWh⁻¹; at 60 14–59% of the pairs install an array (none for seasonal sharing) and the net values are small (static arrays 0.9, SALS 0.3 kUSD); static arrays are the best option up to about 75 USD MWh⁻¹, SALS leads from about 80 (4.9 against 4.3 at 80, 8.3 against 6.3 at 90, 20.7 against 12.3 at 120), and the tuned rule overtakes static arrays only above about 105. The value of control is therefore real but depends on a high price: at 90 USD MWh⁻¹ SALS is worth 1.7 kUSD ha⁻¹ yr⁻¹ over the tuned rule with profit-optimal densities.

Cost of the safety margin. At 90 USD MWh⁻¹, moving from a density designed under the mean response to a floor of 0.92 under that response costs 1.00 kUSD ha⁻¹ yr⁻¹ (17%) under E2 for 18 more points of compliance (70% to 88%) and 1.13 kUSD (20%) under E3 for 14 points (71% to 85%), about 56 and 79 USD ha⁻¹ yr⁻¹ per point. The ensemble rule that requires all other members to meet the floor costs 0.85 and 1.22 kUSD for 19 and 17 points (45 and 74 USD per point), so on this measure it is cheaper than the margin under E2 and similar under E3. At 60 USD MWh⁻¹ a margin has a negative cost, because every module beyond break-even loses money; the economics of the design margin therefore depend on the price level, whereas its effect on compliance does not.

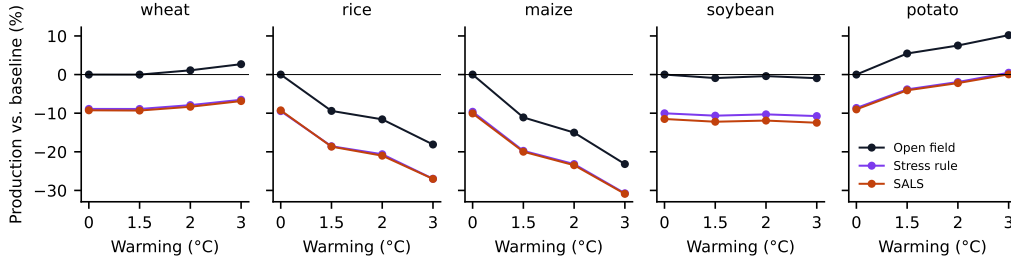


Figure 8: Production relative to the baseline open field (sum over held-out site-seasons) as a function of warming, for the open field, the stress rule and SALS.

3.10. Agrivoltaics under warming

Under warming, the open-field production of the held-out sites (Fig. 8) falls at +3 °C by 23% for maize and 18% for rice, is nearly unchanged for soybean, and rises for wheat (+3%) and potato (+10%). Agrivoltaic production with light-sharing control follows the open field at about 90% of its level, so it inherits the same climate impacts (maize −31%, rice −27% relative to the baseline open field at +3 °C). Agrivoltaics does not protect staple production against warming in this model. The LER of light-sharing systems increases slightly with warming (SALS 1.270 to 1.297; phenology rule 1.236 to 1.265), because the floor allows slightly denser arrays in warmer conditions (GCR 0.173 to 0.177 and electricity 510 to 540 MWh ha^{−1} yr^{−1}). The margin of SALS over the rules does not grow with warming (Table A.17).

3.11. Global potential and what it would take

Every site is evaluated when it is held out (219 of the 400 crop–site pairs, in at least one split), and each crop’s mean electricity per hectare is scaled to its physical area in SPAM 2010 (581 Mha for the five crops, which avoids counting multiple-cropped land twice). Equipping 5% of this area (29.1 Mha) with SALS arrays at the floor would produce 14.4 PWh yr^{−1} under the baseline climate and 16.0 PWh yr^{−1} at +3 °C, with a production-weighted retention of 0.903 and 0.900. The phenology rule would give 13.0 and 14.5 PWh yr^{−1}, and static arrays 6.4 and 6.9 PWh yr^{−1}. For comparison, world electricity generation was about 30 PWh in 2023 (Ember, 2024).

These numbers require large amounts of hardware. SALS at 5% coverage needs 10.6 TWp of modules (11.3 TWp at +3 °C), at a capacity factor of 15.5% (16.2%). World solar PV capacity at the end of 2024 was about 1.9 TW

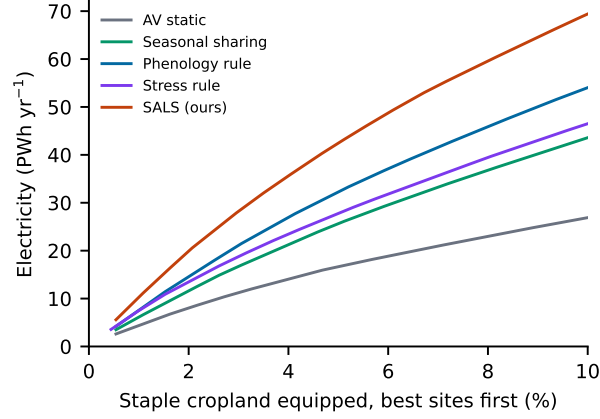


Figure 9: Supply curve under the baseline climate: cumulative electricity when sites are equipped in order of decreasing electricity per hectare, against the share of staple cropland (physical area) equipped. Each controller is evaluated at the density that keeps its mean retention at 0.9.

(IRENA, 2025), so the requirement is about 5.7 times the installed PV fleet. The electricity is also variable, peaks at midday and would be curtailed or stored. The figures are technical potentials, not projections. Ordering sites by electricity per hectare gives a supply curve (Fig. 9): the best 2% of the staple cropland would supply 16 PWh yr⁻¹ and the best 5% 40 PWh yr⁻¹ with SALS. The best tenth of the sites are mostly rainfed maize and soybean sites in the Americas and in southern and eastern Africa (7% mean irrigated fraction against 29% overall), which are also among the regions where the floor is hardest to meet (Section 3.5). Siting by electricity alone would therefore concentrate agrivoltaics where food-security risk is highest.

Maize and soybean, for which the simulated shade response is probably too mild (Section 3.7), account for 53% of the electricity of the upscaling (maize 30%, soybean 23%, wheat 29%, rice 15%, potato 3%). Counting only wheat, rice and potato would give about 7 PWh yr⁻¹ at the same coverage, so the global figure should be read as an upper estimate for about half of its content. By crop, the LER of SALS under the baseline climate is 1.20 for wheat and rice, 1.31 for potato, 1.30 for maize and 1.41 for soybean.

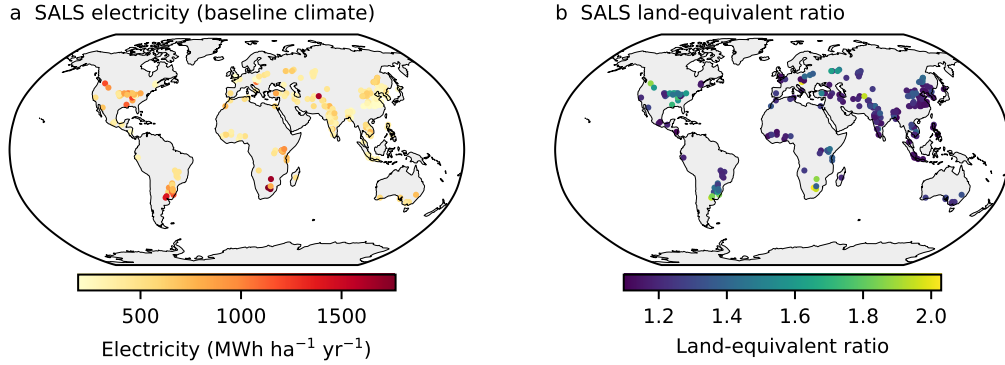


Figure 10: SALS electricity (a) and land-equivalent ratio (b) at the pooled held-out sites (baseline climate, mean of seasons 2015–2019).

837 4. Discussion

838 4.1. What the results establish

839 Four findings stand on consistent evidence. First, sharing light with the
840 crop is the main lever on the electricity that tracker-based agrivoltaic arrays
841 can produce under a yield floor. With static trackers the floor limits arrays
842 to about 6% ground coverage and an LER of 1.07. Any light-sharing control
843 allows 2.0 to 2.9 times the density, and the LER rises to 1.18–1.24 for hand-
844 written rules. Second, the best simple rule follows crop phenology. It shares
845 light across the middle of the season, around flowering and grain filling, and
846 produces electricity at the beginning and the end. It needs only the crop’s
847 thermal time, which is observable on the farm, and it is insensitive to forecast
848 errors, whereas the rule that reacts to heat and drought stress is worse in
849 every climate and degrades with forecast error. Third, the qualitative result
850 does not depend on the weakest part of the model. The simulated shade
851 response is conservative against the field trials we could use (Section 3.6);
852 calibrating it raises the allowed density, and the phenology rule stays the
853 best method, with a gain over static arrays of 0.17, 0.19 and 0.30 LER
854 under the conservative, calibrated and field-fitted responses. However, the
855 response learned from the meta-analysis shows steeper losses for maize and
856 grain legumes than the simulator reproduces (Section 3.7), so the absolute
857 gains are conservative for wheat, rice and potato and probably overestimates
858 for maize and soybean (53% of the upscaled electricity). Fourth, a controller
859 learned through the differentiable simulator is better than the best rule we

could write, by 0.04 in LER against the written rule (0.018 against a rule whose window was tuned on the training seasons, 5.7% more electricity at equal retention), reproducibly over training seeds (range 0.002 in LER), three spatial splits, three canopy-cooling assumptions and four climates, and at 84–86% of the held-out sites. This gain is small compared with the gap between static and dynamic control.

4.2. *What role does artificial intelligence play?*

We tested three ways in which learning could matter, and the answers differ. The first is learning the crop’s response to shade from data. The learned model is useful mainly as a measurement of how little can be learned: a hierarchical model fitted to 428 observations from 58 studies predicts a held-out study of the four staple crop types with an error of 0.22 in relative yield, no better than proportional loss, and the gain over proportional loss (0.43 to 0.31 over all types) comes from crop types that the simulator does not represent. Crop type is the information that transfers, whereas irradiation, climate zone and shade type add nothing measurable and a non-parametric model (gradient-boosted trees) is worse with this number of studies (Section 3.7). Two things follow. The shade response of the staple crops is a distribution with a spread of about 20 percentage points between studies and not a curve, and the process-based simulator does not reproduce the steep losses of maize and legumes seen in the data. Learning the response was limited by the data and not by the model: two studies support the tuber curve, no rice study exists, and 46 panel observations are all leafy vegetables.

The second is calibration through the differentiable simulator. Gradient descent through whole seasons fitted the response of each crop to the learned curves and to the field trials, and it produced an ensemble of plausible responses that reconciles them (Section 3.8). The ensemble informs the design decision: a density chosen under the mean response leaves about 70% of crop–site pairs at the floor if another plausible response holds, against 85–87% for an oracle, and a density chosen to meet the floor in 75% of plausible responses recovers about half of that gap at a cost of 0.04 LER. A simple margin, however, does as well: raising the floor from 0.90 to 0.92 under the mean response reaches 85–88% of the pairs, and at equal compliance it keeps 0.01 (E2) to 0.03 (E3) more LER than the ensemble rule that requires all other members to meet the floor, while the cost per point of compliance is similar. None of the rules is safe outside the ensemble (7–33% of the pairs at the floor under the proportional, calibrated and harsh responses for the

mean design, the margin of 0.92 and the ensemble rules), and only a floor of 0.96 reaches 51–78% at a cost of about 0.2 LER. What the ensemble adds is the size of the margin, not a better rule. The quantile rule is a scenario-type design (Calafiore and Campi, 2006; Campi and Garatti, 2008); with five members it has none of the finite-sample guarantees of that theory, and the comparison with the margin shows that it is not needed in practice. The bottleneck of tracker-based agrivoltaics under a food-security floor is the uncertainty of the crop response that enters the design, and it is data on that response, not larger networks, that would improve the result; the learned curves show a spread of about 20 percentage points between studies. The guarantee holds only within the range of responses that the ensemble spans.

The third is control, and here learning contributes little. The learned controller adds 0.018 LER over a tuned phenology rule (and 0.044 over the written one), and 0.015–0.038 over the written rule under the four alternative responses on one split (21 pairs); a reader who needs a simple, auditable controller can use that rule and obtain most of the electricity gain over static arrays. Neither of the two extensions that we expected to matter under uncertainty did: training on ensembles of responses (E1, E2 and E3) did not help and, for the wider ensembles, did not retain the margin over the written rule within our training budget (one seed, 21 pairs), and neither did giving the controller a farm-measurable signal, the biomass of the crop relative to a reference plot, nor a feedback rule built on that signal. The learned policy also has to be initialised from the phenology rule (Section 4.5). We therefore do not claim that learned control is what makes agrivoltaics work: in this model, the schedule that a learned policy converges to is a smooth function of crop phenology, which a rule can already express.

4.3. *What the results do not establish*

The early hypothesis that stress-aware control, that is, shading the crop on heat and drought days, would be a main source of benefit under warming was not supported. The stress rule performs worse than the phenology rule in every climate, and a learned controller without stress inputs performs as well as one with them. A plausible reason is that in our crop model the yield gain from shading on a hot or dry day is smaller than the loss of light on that day, so stress information does not help. Whether this holds for crop models with more detailed canopy-temperature and heat-stress responses is an open question. Agrivoltaics does not protect staple production against warming in our simulations: light-sharing systems produce about 90% of the

934 open-field yield in every climate and follow the same decline in maize and
 935 rice (Fig. 8).

936 4.4. Food security in practice

937 Holding the historical mean retention at 90% does not hold it in each
 938 season or at each site. 70% of SALS seasons meet the floor, the lower tail is
 939 slightly heavier than for the rules, and production-weighted retention is below
 940 0.9 for soybean (0.888) and maize (0.898) and in several regions. Chance-
 941 constrained design raises season-level compliance to about 92% at a cost of
 942 0.065 in LER, and the ranking of the controllers is preserved. Whichever for-
 943 mulation is chosen, a per-site regulatory floor would need a region- and crop-
 944 specific design step. The upscaled potential (Section 3.11) shows the trade-off
 945 in magnitude. The 14 PWh yr⁻¹ that 5% of the staple cropland could supply
 946 need about 10 TWp of modules, several times the present global fleet, and
 947 the best sites are rainfed maize and soybean areas where production-weighted
 948 retention is lowest.

949 4.5. Limitations

- 950 • *Simulation only.* All results come from a simulator. PV output was
 951 benchmarked against PVGIS (29 sites, mean bias -4.9%), but the crop
 952 response to shade was learned from one meta-analysis data set and
 953 compared with four field trials (Sections 3.6 and 3.7). The simulated
 954 response is conservative with respect to the trials (rice, wheat, potato,
 955 maize) but too mild for maize and soybean relative to the meta-analysis,
 956 where the 40%-shading relative yields are 0.42 and 0.40 against 0.75
 957 and 0.48 in the simulator. The harvest-index sensitivities h are assump-
 958 tions, and the sensitivity analysis uses rule controllers on a subsample
 959 of 22 sites.
- 960 • *Canopy cooling.* The effect of shade on canopy temperature was repre-
 961 sented by one coefficient κ , applied to a heat-stress function that was
 962 calibrated on air temperature (Siebert et al., 2014; Webber et al., 2018).
 963 The ranking of the controllers is robust to κ between 0 and 2°C, but
 964 the absolute electricity changes by about $\pm 8\%$. Nocturnal warming
 965 under modules and rain redistribution are not modelled.
- 966 • *Crop model.* SIMPLE has a fixed harvest index and linear radiation-
 967 use efficiency. More detailed models with grain-number formation and
 968 canopy energy balance may favour different schedules.

- 969 • *Climate scenarios.* Warming was applied as uniform temperature shifts
970 with fixed precipitation, radiation and relative humidity, and with un-
971 changed cultivars and sowing dates. It is not a bias-adjusted climate
972 projection.
- 973 • *Initialisation and state information.* The network is initialised from the
974 phenology rule, because training from random weights collapsed to full
975 sharing. The gain of SALS is therefore the gain of fine-tuning that rule,
976 and we cannot say whether a better optimiser or a richer policy class
977 would find larger gains. The controller uses the simulated state of the
978 crop and soil (thermal time, canopy interception, soil water) without
979 measurement error, and the forecast errors we tested apply to weather
980 only.
- 981 • *Economics and deployment.* The economic analysis (Section 3.9) uses
982 assumed prices and costs, no grid, curtailment, storage or subsidy
983 model, and does not value food security; its break-even prices move
984 with the capex and the discount rate (Fig. 7). The value of midday
985 electricity, curtailment, storage, structural cost and farmer acceptance
986 (Trommsdorff et al., 2021) will determine the realisable share. The av-
987 erage row pitch at the chosen densities (about 13 m for 2.2-m modules)
988 leaves room for machinery, but the densest designs (GCR up to 0.5)
989 have a pitch of only 4.4 m.

990 5. Conclusions

991 Light sharing, the daily choice between maximum electricity and letting
992 light through to the crop, is the main lever of tracker-based agrivoltaics under
993 a food-security floor. In a differentiable simulator of trackers, PV output,
994 crop light, microclimate and crop growth, evaluated on held-out crop-site
995 pairs (92 on the main split, 76 distinct pairs pooled over three splits for the
996 controller and design comparisons) with the array density of each site chosen
997 from its own history, light-sharing rules allowed 2.0 to 2.9 times the module
998 density of static arrays at a mean yield retention of 90%. They raised the
999 land-equivalent ratio from 1.07 to 1.18–1.24, and a rule that follows crop
1000 phenology and needs no weather forecast was the best of them. The simu-
1001 lated shade response is conservative against the field trials we could use for
1002 rice, wheat and potato, and calibrating it raised the gains without chang-
1003 ing the ranking of methods; for maize and soybean, which supply 53% of

1004 the upscaled electricity, the meta-analysis indicates larger losses than the
 1005 simulator reproduces whereas one maize trial indicates smaller ones, so the
 1006 gains for these crops are uncertain and possibly overestimated. The crop's
 1007 response to shade, the least certain input, was learned from 428 field obser-
 1008 vations: a hierarchical model predicted a held-out study of the staple crop
 1009 types no better than proportional loss (error 0.22), which shows the size of
 1010 the between-study spread, and gradient descent through the simulator turned
 1011 it into an ensemble of plausible responses. A learned controller exceeded the
 1012 written phenology rule by 0.044 LER and a rule with a window tuned on the
 1013 training seasons by 0.018 (95% interval 0.012–0.024), by following a smooth
 1014 phenology-dependent schedule that a rule can express; training it on ensem-
 1015 bles of shade responses or giving it a farm-measurable biomass signal did
 1016 not help, and with the wider ensembles it did not retain its margin over the
 1017 written rule in our training budget (one seed, 21 pairs). The decision that
 1018 mattered was the design. A density chosen under the mean response left
 1019 about 70% of crop–site pairs at the floor when another plausible response
 1020 held (oracle 85–87%); choosing the density so that the floor holds in 75%
 1021 of the plausible responses recovered about half of the gap at a cost of 0.04
 1022 LER, but a floor raised by 0.02 under the mean response was as good as or
 1023 slightly better at equal compliance, and no rule was safe outside the ensem-
 1024 ble. The ensemble is therefore useful to size a safety margin, not to replace
 1025 it. Under illustrative costs, light sharing breaks even with the open field at
 1026 73 (learned controller) to 92 USD MWh^{−1}; a conventional plant replacing the
 1027 crop stays more profitable at every price, so agrivoltaics depends on a pre-
 1028 mium, land-use rules or a value for food production that markets do not give.
 1029 The yield floor is met on average but not in every season or region, and the
 1030 electricity potential of 5% of staple cropland (about 14 PWh yr^{−1}) requires
 1031 about 10 TWp of modules. Future work should validate the crop response
 1032 to shade under trackers with more field data, notably for maize and legumes
 1033 where the meta-analysis and the trials disagree, extend the crop model to
 1034 grain-number formation and canopy energy balance, add costs and grid con-
 1035 straints, and pursue controllers that guarantee the food-security floor per site
 1036 and per season.

1037 **CRedit authorship contribution statement**

1038 **Quang-Vinh Dang:** Conceptualization, Methodology, Software, Formal
 1039 analysis, Investigation, Data curation, Visualization, Writing – original draft,

1040 Writing – review & editing.

1041 Declaration of competing interest

1042 The author has no known competing financial interests or personal re-
1043 lationships that could have appeared to influence the work reported in this
1044 paper.

1045 Data and code availability

1046 All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al.,
1047 2020), the GGCM Phase 3 crop calendar (Jägermeyr et al., 2021), NASA
1048 POWER daily weather (Sparks, 2018) and the SIMPLE crop parameters
1049 (Zhao et al., 2019). The code of the simulator, of SALS and of all experi-
1050 ments, together with the site list, is available from the author’s public reposi-
1051 tory (<https://github.com/vinhqdang/Artificial-Intelligence-in-Agriculture>).

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1055 Appendix A. Statistical comparison

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Table 2: Held-out performance (92 test sites, seasons 2015–2019, canopy cooling $\kappa = 1^\circ\text{C}$). The density (GCR) of each site is the largest value on the grid 0.05, 0.075, ..., 0.5 at which the method’s mean retention in the site’s 2001–2014 seasons is ≥ 0.9 . E : electricity ($\text{MWh ha}^{-1} \text{ yr}^{-1}$); e : electricity relative to a conventional PV plant; r : mean yield retention; r_{prod} : production-weighted retention; C_s and C_{site} : share of seasons with $r \geq 0.9$ and share of sites whose test-season mean $r \geq 0.9$ (%); $\text{LER} = r + e$.

Method	GCR	E	e	r	r_{prod}	C_s	C_{site}	LER
<i>Baseline</i>								
AV static	0.06	217	0.16	0.917	0.914	93	96	1.07
Seasonal sharing	0.17	397	0.28	0.904	0.902	69	72	1.19
Phenology rule	0.17	465	0.33	0.904	0.902	69	75	1.24
Stress rule	0.13	387	0.28	0.907	0.907	76	82	1.18
SALS (ours)	0.17	510	0.36	0.905	0.902	70	74	1.27
<i>+1.5 °C</i>								
AV static	0.06	221	0.16	0.916	0.914	93	92	1.08
Seasonal sharing	0.16	393	0.28	0.905	0.903	72	74	1.19
Phenology rule	0.17	478	0.34	0.905	0.902	68	75	1.25
Stress rule	0.12	377	0.27	0.908	0.908	73	77	1.18
SALS (ours)	0.17	518	0.37	0.906	0.903	68	75	1.28
<i>+2 °C</i>								
AV static	0.06	223	0.16	0.916	0.914	92	93	1.08
Seasonal sharing	0.17	394	0.28	0.905	0.904	72	76	1.19
Phenology rule	0.17	483	0.35	0.905	0.903	69	74	1.25
Stress rule	0.12	370	0.26	0.909	0.909	75	78	1.17
SALS (ours)	0.17	521	0.38	0.907	0.904	69	75	1.28
<i>+3 °C</i>								
AV static	0.06	231	0.17	0.915	0.914	88	89	1.08
Seasonal sharing	0.17	397	0.29	0.906	0.904	69	75	1.19
Phenology rule	0.18	499	0.36	0.905	0.902	67	72	1.26
Stress rule	0.11	366	0.26	0.908	0.908	74	77	1.17
SALS (ours)	0.18	540	0.39	0.907	0.904	68	75	1.30

Table 3: LER of the controllers at different yield-retention floors ρ (main split; the historical design step is repeated for each floor).

Method	Baseline climate				+3 °C			
	0.80	0.85	0.90	0.95	0.80	0.85	0.90	0.95
AV static	1.159	1.115	1.072	1.011	1.174	1.128	1.083	1.013
Stress rule	1.380	1.282	1.182	1.077	1.364	1.269	1.172	1.071
Phenology rule	1.467	1.356	1.236	1.108	1.513	1.395	1.265	1.122
SALS (trained at $\rho = 0.9$)	1.505	1.392	1.270	1.141	1.553	1.432	1.297	1.154
SALS conditioned on ρ	1.504	1.391	1.269	1.141	1.553	1.431	1.296	1.154

Table 4: Rules and the learned controller pooled over the three spatial splits (76 distinct crop-site pairs, every third held-out pair; baseline climate; main shade response; history-based design on a GCR grid with step 0.05, coarser than the 0.025 grid of the main analysis). The tuned phenology rule shares light while thermal time lies between 0.15 and 1.0 of the season requirement (window chosen on training-block seasons of split 0; the written rule uses 0.05 to 1.0). The optimised ramp schedule has five parameters (start 0.17, end 0.93, ramp widths 0.02, level 0.99) found by CMA-ES on the same training seasons.

Method	GCR	E/E_{PV}	Retention	Seasons ≥ 0.9 (%)	Pairs ≥ 0.9 (%)	LER
AV static	0.059	0.153	0.922	95	96	1.076
Seasonal sharing	0.151	0.249	0.915	84	91	1.164
Stress rule	0.118	0.255	0.918	83	88	1.173
Phenology rule	0.154	0.292	0.916	85	91	1.207
Tuned phenology rule	0.161	0.317	0.916	84	89	1.233
Optimised ramp schedule	0.155	0.322	0.916	83	87	1.238
SALS	0.161	0.335	0.916	83	87	1.251

Table 5: Simulated relative yield at the shading of published field trials (static arrays, training-block sites). Observed values: rice, single trial; wheat and potato, the worse of the two seasons (the better season gave +2.7% and +11%); maize, mean of the three seasons of the dynamic system (DAV) and single values for the other two systems. The same trials enter the calibration of Section 2.6, so the comparison is not independent.

Trial	Shading	Observed	Conservative	Calibrated
Rice, Gonocruz et al. (2021)	0.25	0.84	0.77	0.79
Wheat, Weselek et al. (2021)	0.30	0.81	0.66	0.74
Potato, Weselek et al. (2021)	0.30	0.82	0.72	0.72
Maize, Ramos-Fuentes et al. (2023) (DAV)	0.35	0.83	0.70	0.70
Maize, Ramos-Fuentes et al. (2023) (half)	0.30	0.91	0.75	0.75
Maize, Ramos-Fuentes et al. (2023) (full)	0.50	0.70	0.54	0.54

Table 6: Sensitivity of the deployable light-sharing results to the crop shade response (rule controllers, baseline climate, floor $\rho = 0.9$, split 0, every fourth held-out site, 22 sites). Conservative: proportional response (main analysis); calibrated: radiation-use compensation fitted to Laub et al. (2022); field-fitted: fitted to the field trials, maize harvest index unaffected.

Shade response	Method	GCR	E/E_{PV}	Retention	LER
conservative	AV static	0.06	0.15	0.92	1.07
	Phenology rule	0.17	0.33	0.90	1.23
	Seasonal sharing	0.16	0.28	0.90	1.18
	Stress rule	0.11	0.24	0.91	1.15
calibrated	AV static	0.07	0.18	0.91	1.09
	Phenology rule	0.19	0.37	0.90	1.28
	Seasonal sharing	0.19	0.32	0.90	1.22
	Stress rule	0.13	0.29	0.91	1.20
field-fitted	AV static	0.16	0.41	0.91	1.32
	Phenology rule	0.37	0.70	0.92	1.62
	Seasonal sharing	0.37	0.61	0.92	1.53
	Stress rule	0.29	0.63	0.90	1.53

Table 7: Predicting relative yield under shade from 428 observations of 58 studies (Laub et al., 2022): leave-one-study-out cross-validation (every study held out once; observations with $s > 0$). Covariates: irradiation, climate zone, shade type, intercropping.

Model	RMSE	MAE	Median AE	Mean MAE per study
Proportional loss, $y = 1 - s$	0.430	0.305	0.210	0.224
One curve for all crops	0.372	0.265	0.193	0.208
Curve per crop type, no pooling	0.313	0.208	0.141	0.165
Curve per crop type, partial pooling	0.312	0.210	0.150	0.162
Partial pooling + covariates	0.311	0.206	0.148	0.161
Gradient-boosted trees	0.321	0.219	0.162	0.184

Table 8: Leave-one-study-out prediction error (RMSE of relative yield, observations with $s > 0$) by crop type: proportional loss, hierarchical curve with partial pooling, and gradient-boosted trees. The last columns give the difference hierarchical minus proportional with a 95% cluster-bootstrap interval over studies (negative: hierarchical better).

Group	Obs./studies	Proportional	Hierarchical	Boosted t
C3 cereals	30/10	0.214	0.187	0.205
Maize	40/10	0.225	0.235	0.248
Grain legumes	58/14	0.215	0.225	0.234
Tubers/root crops	10/2	0.250	0.254	0.257
Four crop types used by the simulator	138/30	0.220	0.222	0.234
Other types (berries, forages, fruits, vegetables)	201/29	0.528	0.361	0.369
All types	339/58	0.430	0.312	0.321

Table 9: Relative yield at 20% and 40% shading: data-driven model (median and 80% bootstrap interval over studies; crop-type curve, covariates at their reference values: mean irradiation, temperate zone, no solar-panel shade, no intercropping) and simulator under the two calibrated ensembles (mean of members; E1 fitted to hand-read curves, E2 fitted to the data-driven curves). The data-driven curve of rice is that of C3 cereals (no rice in the data set). Observations and studies: C3 cereals 41/10, maize 51/10, grain legumes 76/14, tubers 12/2.

Crop	Shading	Data-driven	Simulator, E1	Simulator, E2	Proportional
Wheat	20%	0.93 [0.88, 0.96]	0.88	0.92	0.80
Wheat	40%	0.72 [0.59, 0.78]	0.70	0.77	0.60
Rice	20%	0.93 [0.88, 0.96]	0.85	0.90	0.80
Rice	40%	0.72 [0.59, 0.78]	0.66	0.73	0.60
Maize	20%	0.73 [0.58, 0.86]	0.86	0.90	0.80
Maize	40%	0.42 [0.27, 0.59]	0.69	0.75	0.60
Soybean	20%	0.71 [0.65, 0.79]	0.77	0.74	0.80
Soybean	40%	0.40 [0.34, 0.53]	0.53	0.48	0.60
Potato	20%	0.93 [0.92, 0.97]	0.88	0.92	0.80
Potato	40%	0.71 [0.67, 0.83]	0.73	0.79	0.60

Table 10: Shade-response parameters of ensemble E2, calibrated by gradient descent through the simulator to the data-driven curves and the field trials (12 randomised fits; mean $\pm$ s.d.): radiation-use compensation c and multiplier h on the flowering-window harvest-index penalty (1 = assumed value).

Crop	c	h multiplier
Wheat	0.91 ± 0.31	0.74 ± 0.53
Rice	0.61 ± 0.25	0.83 ± 0.33
Maize	0.14 ± 0.11	0.16 ± 0.13
Soybean	0.05 ± 0.09	1.66 ± 0.38
Potato	0.62 ± 0.20	0.90 ± 0.50

Table 11: Land-equivalent ratio of rules and learned controllers under four true shade responses, with the array density designed under the true response (split 0, baseline climate, floor $\rho = 0.9$, every sixth held-out pair, 21 pairs, one training seed; season means, so differences differ slightly from the pair-level intervals in the text). Robust controllers are trained on random members of ensemble E1, E2 or E3 (adaptive: with the measured biomass ratio). The proportional (conservative) and harsh responses lie outside the ensembles.

Method	Proportional	Calibrated	Field-fitted	Harsh
AV static	1.052	1.059	1.254	1.037
Seasonal sharing	1.133	1.156	1.385	1.097
Stress rule	1.129	1.160	1.406	1.086
Feedback rule	1.139	1.164	1.392	1.089
Phenology rule	1.164	1.190	1.456	1.122
SALS (trained on proportional response)	1.178	1.209	1.489	1.132
SALS-robust, E1	1.179	1.204	1.491	1.127
SALS-adaptive, E1	1.176	1.202	1.495	1.124
SALS-robust, E2	1.165	1.192	1.483	1.126
SALS-robust, E3	1.163	1.191	1.481	1.121
SALS-adaptive, E3	1.164	1.191	1.475	1.118

Table 12: Array design under shade-response uncertainty, phenology rule, three spatial splits pooled (76 distinct crop-site pairs, every third held-out pair). Truth: each of five members of the ensemble in turn; the mean response and the member-based rules use the other four members (leave-one-out, so the truth never enters the design). E2: ensemble calibrated to the learned curves and field trials; E3: same procedure with target errors of 0.2 on the learned curves. Pairs: share with mean retention ≥ 0.9 ; shortfall: mean of $\max(0, 0.9 - \text{retention})$ per pair.

Design	E2		LER	E3		
	Pairs ≥ 0.9 (%)	Shortfall		Pairs ≥ 0.9 (%)	Shortfall	
oracle (true response)	87	0.0025	1.486	85	0.0029	1
mean response	70	0.0062	1.493	71	0.0093	1
floor met in 50% of other members	65	0.0084	1.510	67	0.0125	1
floor met in 75% of other members	78	0.0045	1.462	79	0.0083	1
floor met in all other members	89	0.0019	1.412	88	0.0032	1
floor 0.92 under mean response	88	0.0022	1.426	85	0.0051	1
floor 0.94 under mean response	97	0.0007	1.345	92	0.0027	1
floor 0.96 under mean response	99	0.0003	1.251	96	0.0015	1

Table 13: Design rules outside the ensemble: share of crop–site pairs (%) with mean retention ≥ 0.9 under four fixed responses (P proportional, C calibrated to the meta-analysis, F field-fitted, H harsh), phenology rule, three splits pooled (76 distinct pairs). Designs use the whole ensemble E2 or E3 (five members and their mean); the oracle uses the true response. LER in the last column is the mean over the four responses.

Design	E2				E3				LER	
	P	C	F	H	P	C	F	H	E2	E3
oracle (true response)	91	89	87	84	91	89	87	84	1.283	1.283
mean response	13	13	68	12	13	17	79	7	1.435	1.448
floor met in 75% of members	16	16	80	12	22	25	93	13	1.384	1.397
floor met in all members	17	17	80	12	29	32	93	16	1.348	1.336
floor 0.92 under mean response	17	20	86	12	29	33	91	20	1.376	1.399
floor 0.94 under mean response	45	47	97	34	45	45	97	29	1.303	1.345
floor 0.96 under mean response	63	78	99	54	58	63	99	51	1.220	1.264

Table 14: Assumed economic parameters (illustrative, not data). Ranges used in the sensitivity analysis are in parentheses.

Parameter	Value
Electricity price	60 USD MWh ⁻¹ (30–120)
Capex, AV single-axis tracker plant	1.0 USD Wp ⁻¹ (0.7–1.5)
Capex, conventional tracker plant	0.75 USD Wp ⁻¹
Discount rate, lifetime, O&M	6% (4–8%), 25 years, 1.5% of capex per year
Installed capacity	2.1 MWp per hectare of module area share g (0.21 kWp m ⁻²)
Crop price per tonne of dry matter	wheat 290, rice 400, maize 250, soybean 520, potato 1000 USD (0.

Table 15: Illustrative annual economics per hectare (76 crop–site pairs, three splits, main shade response, baseline climate; parameters in Table 14). Net value is relative to the open field (crop costs unchanged) in kUSD ha⁻¹ yr⁻¹ at three electricity prices; break-even is the price at which the net value is zero. The conventional plant replaces the crop.

Method	E (MWh)	Capex annuity	Crop loss	Net at 60	Net at 90	Net at 120
AV static	200	11.5	0.27	0.3	6.3	12.3
Seasonal sharing	326	29.6	0.31	-10.3	-0.5	9.2
Stress rule	336	23.2	0.30	-3.3	6.8	16.9
Phenology rule	382	30.1	0.30	-7.5	4.0	15.4
Tuned phenology rule	415	31.6	0.31	-7.0	5.5	17.9
SALS	438	31.6	0.31	-5.6	7.6	20.7
Conventional PV plant	1306	58.7	3.60	16.0	55.2	94.4

Table 16: One-at-a-time sensitivity of the net value (kUSD ha⁻¹ yr⁻¹, relative to the open field) at 90 USD MWh⁻¹, densities designed for the floor as in the main analysis. The capex of the conventional plant is 0.75/0.70 of that of the agrivoltaic plant in the capex rows.

Parameter	Value	Static arrays	Tuned rule	SALS	Conventional plant
Base		6.3	5.5	7.6	55.2
AV capex (USD/Wp)	0.7	9.7	14.9	17.1	72.8
AV capex (USD/Wp)	1.5	0.6	-10.3	-8.2	25.8
Discount rate	4%	8.0	10.3	12.4	64.1
Discount rate	8%	4.4	0.2	2.4	45.4
Crop prices	0.5×	6.4	5.6	7.7	57.0
Crop prices	2.0×	6.0	5.2	7.3	51.6
Electricity price	60	0.3	-7.0	-5.6	16.0
Electricity price	120	12.3	17.9	20.7	94.4

Table A.17: Paired comparison of SALS with each baseline on the 92 held-out sites of the main split (unit of analysis: site mean over test seasons). ΔLER : mean difference (SALS minus baseline) with cluster-bootstrap 95% CI; p : two-sided Wilcoxon signed-rank test; % better: share of sites where SALS has the higher LER; Δr : difference in mean yield retention with its Wilcoxon p -value.

Climate	Baseline	ΔLER [95% CI]	p	% better	Δr	p	Sites
Baseline	AV static	+0.205 [+0.176, +0.236]	$< 10^{-4}$	100	-0.011	$< 10^{-4}$	92
Baseline	Seasonal sharing	+0.088 [+0.068, +0.109]	$< 10^{-4}$	97	+0.001	0.922	92
Baseline	Phenology rule	+0.038 [+0.025, +0.053]	$< 10^{-4}$	84	+0.001	0.723	92
Baseline	Stress rule	+0.096 [+0.071, +0.124]	$< 10^{-4}$	93	-0.002	0.055	92
+1.5 °C	AV static	+0.208 [+0.180, +0.237]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+1.5 °C	Seasonal sharing	+0.095 [+0.077, +0.114]	$< 10^{-4}$	98	+0.001	0.812	92
+1.5 °C	Phenology rule	+0.033 [+0.022, +0.045]	$< 10^{-4}$	86	+0.002	0.855	92
+1.5 °C	Stress rule	+0.108 [+0.083, +0.134]	$< 10^{-4}$	98	-0.001	0.041	92
+2 °C	AV static	+0.209 [+0.181, +0.238]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+2 °C	Seasonal sharing	+0.097 [+0.079, +0.115]	$< 10^{-4}$	97	+0.001	0.876	92
+2 °C	Phenology rule	+0.032 [+0.022, +0.044]	$< 10^{-4}$	85	+0.002	0.496	92
+2 °C	Stress rule	+0.115 [+0.089, +0.142]	$< 10^{-4}$	98	-0.003	0.016	92
+3 °C	AV static	+0.219 [+0.190, +0.250]	$< 10^{-4}$	100	-0.009	$< 10^{-4}$	92
+3 °C	Seasonal sharing	+0.108 [+0.089, +0.129]	$< 10^{-4}$	96	+0.001	0.935	92
+3 °C	Phenology rule	+0.034 [+0.022, +0.046]	$< 10^{-4}$	86	+0.002	0.330	92
+3 °C	Stress rule	+0.131 [+0.102, +0.162]	$< 10^{-4}$	98	-0.002	0.142	92